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ORIGINAL PAPER

# Design and optimization analysis of a lead-free perovskite solar cell based on copper (I) iodide hole transport layer

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## Abstract

Photovoltaic technology has gained wide acceptance because of its potential to mitigate climate change while offering pathways to reduce carbon footprint and inspiring renewable energy access and uptake. Herein, we investigate the performance of a solar cell configuration, FTO/$TiO_2$/$CH_3NH_3SnI_3$/CuI/Pd, by device simulation using solar cell capacitance device simulator (SCAPS-1D). The study analyses the impact of temperature, absorber layer thickness and buffer layers on the electrical and optoelectronic outcomes of the model cell. The simulation reveals optimal power conversion efficiency (PCE) of 23.98%, $V_{oc}$ of 0.8762, modest fill factor (FF) of 75.27% and remarkable photocurrent density ($J_{sc}$) of 34.33 mA/$cm^2$. The selection of palladium as the preferred metal back contact is supported by its high work function, stability in ambient conditions, affordability compared to gold and low resistivity. CdS, distinguished by its remarkable PCE of 24.04%, emerges as the most promising buffer material for the model cell. Increased temperature from 260 to 500 K did not affect the electrical parameters significantly indicating that the model cell with CuI as the hole transport layer (HTL) has a robust temperature tolerance and hence stable in the design of perovskite cell modules based on CuI. The optimal density of defects for the material was generally found to be $10^{15}$ $cm^{-3}$ whereas the optimal density for the acceptor was $10^{16}$ $cm^{-3}$. These findings highlight the feasibility of the model cell design, characterized by use of eco-friendly materials, relatively affordable materials, low toxicity, and impressive PCE, thus inspiring potential fabrication and commercialization.

**Keywords** Interface materials · Perovskite · Affordable materials · Eco-friendly · SCAPS-1D

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## 1 Introduction

The desire to decarbonise the global energy system has paved the way for renewable energy alternatives such as solar energy, which is more dependable, cheap, inexhaustible, and sustainable; however, there are significant challenges associated with stability, durability, and scalability with photovoltaic technologies. By carefully optimizing interfacial properties of solar cell architectures, these bottle-necks can be addressed, consequently lowering the cost of fabrication while improving the performance and longevity of the solar cell designs. This study presents a thorough examination of performance indicators for a perovskite solar cell (PSC) based on copper (I) iodide hole transport layer. In recent years, PSCs have emerged as leading candidates in the renewable photovoltaic (PV) energy landscape (Bhattarai et al. 2023), and it has become necessary to investigate clean, efficient, sustainable, and renewable energy sources to replace conventional fossil fuels which are potential precursors for environmental degradation, climate change and climate action (Olabi and Abdelkareem 2022). The exceptional performance of PSCs in recent years has brought into sharp focus the use of photovoltaic (PV) devices to harness solar energy (Miyasaka and Jena 2022). Due to the exceptional inherent qualities of perovskite materials and the continual improvement of each functional layers, the PCE of cutting-edge PSCs has so far exceeded 25% (Beard and Cole 2022).

This study addresses crucial bottlenecks in the PV technology such as the toxicity of lead and device affordability based on material selection. By focusing on lead-free metal halide perovskites, this investigation marks a significant contribution in sustainable solar energy solutions (López-Fernández et al. 2024). The use of numerical simulation with the SCAPS-1D offers innovative insights into the electrical properties of the model cell under advanced weathering conditions (Njema et al. 2024a). Factors affecting cell efficiency, such as temperature, absorber layer thickness, and metal back contact, is treated in detailed (Bimli et al. 2023). The choice of Pd as the metal back contact underscores the careful design aimed at maximizing efficiency and stability. Overall, this work demonstrates the feasibility of lead-free PSCs, offering environmentally friendly solutions with enhanced efficiency. These advancements hold promise for commercial production of the proposed cell device, highlighting a new dawn in sustainable solar energy technologies. Recent developments in PSCs have led to significant gains in their efficiency, stability, and scalability. Advances in material science, device design, and manufacturing techniques have enabled PSCs to achieve PCEs exceeding 25% (Matondo et al. 2021). These improvements are driven by a deeper understanding of material properties and their interactions within the device, which enhances light absorption and charge transport.

Accordingly, tin-based PSCs can compete favourably with lead-based PSCs. Methylammonium tin iodide ($MASnI_3$) - ($CH_3NH_3SnI_3$) material has recently been used as an absorber layer because of its high absorption capability although it suffers from temperature instability issues (Sajid et al. 2023a). Tin is used as a replacement for Pb because it is less toxic and eco-friendly (Luo et al. 2022; Pandey et al. 2023). The PV industry has recently seen the emergence of perovskite-based solar cells as promising technology (Muhammad et al. 2022). The majority of the perovskite material comprises Pb, which has hazardous properties and restricts its industrial applications (Bhattarai et al. 2022). This work discusses the simulation of a lead-free perovskite solar cell using a $CH_3NH_3SnI_3$ absorber layer because of its low band gap, high visible absorption coefficient and non-toxic composition.

Copper (I) iodide (CuI) based PSCs have emerged as a promising alternative to traditional lead-halide perovskites, offering the potential for enhanced environmental sustainability and reduced toxicity (Aliyaselvam et al. 2023), and has emerged as a highly promising HTL in PSCs due to its favourable conductivity, stability, and low cost (Matondo et al. 2021). Experimental results have shown PSCs using CuI as the HTL achieving PCEs in the range of 15 to 20%, depending on the perovskite composition and device configuration, with improved stability under environmental stresses such as humidity and temperature (Kung et al. 2018). Simulations, particularly with tools like SCAPS-1D, have reported higher PCEs, reaching up to 20% in optimized cell designs (Mandadapu et al. 2018). These studies highlight CuI's effectiveness in enhancing charge extraction and reducing recombination losses at the interface, making it a strong contender to replace traditional HTLs such as Spiro-OMeTAD. However, the potential for further optimization is considerable (Wang et al. 2023). CuI-based PSCs have also demonstrated improved stability, particularly in terms of resistance to moisture and thermal degradation (Jing et al. 2023). This enhanced stability is attributed to the stronger Cu–I bonds and lower ionic conductivity compared to lead-based alternatives (Das et al. 2023).

Another key advantage of CuI-based PSCs is the ability to tune their bandgap by alloying with other halides (Uzagare and Chavan 2023), such as bromine (Br) or chlorine (Cl), enabling optimization for better solar cell performance. Additionally, the crystal structure of CuI, whether cubic or hexagonal, significantly influences its electronic properties and device efficiency, with studies investigating how these phases impact overall performance and stability (Ma et al. 2023). Understanding the defect chemistry of CuI is crucial for controlling its electronic characteristics and minimizing non-radiative recombination, which affects the efficiency of the solar cells (Cui et al. 2023). Researchers have also focused on improving the interfaces between CuI and the charge-transport layers to reduce charge recombination, which is key to enhancing device performance.

In this study, $n{-}i{-}p$ structure PSC with CuI as HTL was numerically simulated. CuI is a promising hole transport layer because of its high hole mobility, good stability, excellent band alignment and has been found to be compatible with perovskite solar cell structures (Shi et al. 2021; Srivastava et al. 2022). Moreover, CuI has a wide optical bandgap of 3.1 eV and high hole mobility $> 40\ \mathrm{cm^2V^{-1}\ s^{-1}}$ which facilitates effective charge transport within the device, large room-temperature exciton binding energy of 62 meV, which therefore enhances the optoelectronic throughput of the perovskite cell architecture (Shi et al. 2021; Peng et al. 2020; Yang et al. 2016). Generally, inorganic HTLs are low cost and have high stability and mobility, and hence better alternatives compared to organic HTLs (Shi et al. 2021). Remarkably, CuI shows potential as a HTL in PSCs due to several advantageous characteristics in contrast to other substances. Its energy levels are well-matched with those of perovskite layers, enhancing efficient charge extraction and minimizing recombination losses. Furthermore, CuI is both cost-effective, making it a feasible option for large-scale production and commercialization of PSCs.

Further, CuI has robust compatibility with solution-based deposition methods which allows for straightforward and scalable fabrication processes, potentially reducing manufacturing expenses. CuI can be deposited on a perovskite thin-film using techniques such as solution processing or thermal evaporation (Kaushik et al. 2017). In the solution-based approach, CuI is dissolved in a solvent like acetonitrile to form a solution, which is then spin-coated onto the perovskite layer, ensuring an even distribution (Er et al. 2020). The film is subsequently annealed at a moderate temperature to remove the solvent, resulting in a crystalline CuI layer. Alternatively, thermal evaporation involves vaporizing CuI under vacuum conditions and depositing it onto the perovskite film, allowing for precise control

over the layer's thickness and uniformity. Both methods yield a smooth, defect-free CuI HTL that facilitates efficient hole extraction, minimizes recombination, and improves the performance of perovskite solar cells. Overall, CuI emerges as a promising HTL material in PSCs, providing a promising blend of performance, affordability, and sustainability compared to alternative materials. Moreover, CuI has the potential to reduce energy losses at interface between HTL and the photoactive layer which therefore enhances charge extraction efficiency and inhibits charge recombination which leads to a higher PCE (Noorasid et al. 2023).

In comparison to Spiro-OMeTAD and other expensive organic HTLs, CuI offers a more affordable solution while maintaining good performance (Bidikoudi and Kymakis 2019). Its deposition can be achieved at relatively low temperatures, minimizing the risk of thermal damage to the perovskite layer and other components. This property is particularly advantageous in the fabrication of PSCs, where maintaining the integrity of the perovskite layer is critical to overall performance. Another reason for selecting CuI as the HTL is its eco-friendly characteristics. Unlike other inorganic materials that may release toxic ions, CuI poses minimal environmental and toxicity risks, making it a more sustainable choice for solar cell applications. While organic HTLs like Spiro-OMeTAD are easier to deposit using simple techniques like spin coating, they are overshadowed by issues related to stability, cost, and conductivity (Nath et al. 2022). In contrast, although inorganic HTLs, including CuI, often require more complex deposition methods, they provide superior stability, better charge transport properties, and enhanced overall device operational performance. This trade-off between deposition simplicity and performance justifies the selection of CuI as the HTL in perovskite solar cells aimed at achieving both efficiency and long-term durability.

Generally, the choice of HTL in PSCs is crucial for determining the device's efficiency, stability, and overall performance (Njema et al. 2024b). Both organic and inorganic HTLs are commonly used, each with its own set of advantages and drawbacks. Organic HTLs, such as Spiro-OMeTAD (2,2′,7,7′-Tetrakis[N,N-di(4-methoxyphenyl) amino]-9,9′-spirobifluorene), PEDOT:PSS (Poly(3,4-ethylenedioxythiophene) polystyrene sulfonate), and PTAA (Poly(triarylamine)), are widely employed due to their relatively simple deposition methods and good compatibility with perovskite materials (Song et al. 2023). However, these organic materials suffer from several significant limitations. One of the major challenges is their low stability. Organic HTLs, particularly Spiro-OMeTAD, tend to degrade over time when exposed to environmental factors such as moisture, oxygen, and temperature variations (Rombach et al. 2021), which results in a reduced operational lifespan for PSCs. Another drawback of organic HTLs is the complexity involved in doping processes, which are necessary to enhance their conductivity (Schloemer et al. 2019). For instance, Spiro-OMeTAD requires doping with additives such as lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI) to increase its hole mobility (Seo et al. 2021). However, the introduction of dopants can introduce long-term stability issues, as the dopants may migrate or crystallize over time, negatively impacting the device's performance. Further, organic HTLs, particularly Spiro-OMeTAD, are relatively expensive, and the synthetic processes used to produce them are complex and costly (Rombach et al. 2021). Organic HTLs also tend to have lower electrical conductivity compared to their inorganic counterparts, which can result in resistive losses and reduced efficiency in the solar cell (Pitchaiya et al. 2020).

Inorganic HTLs, such as NiOx (Nickel Oxide), CuSCN (Copper(I) Thiocyanate), and CuI, offer improved stability and better electrical performance but come with their own challenges (Arumugam et al. 2021). One of the key issues with inorganic HTLs is the

complexity of the deposition processes. In contrast to organic HTLs, which can be easily deposited through solution-based techniques like spin coating, inorganic HTLs often require more advanced deposition methods, such as thermal evaporation, sputtering, or chemical vapor deposition (CVD) (Li et al. 2022). These methods tend to be more expensive, require specialized equipment, and may involve higher processing temperatures, which can potentially damage the perovskite layer or other components of the solar cell. In addition to the complexity of deposition, some inorganic HTLs suffer from interfacial mismatch with the perovskite layer, which can lead to increased charge recombination at interfaces (Njema et al. 2025). This recombination reduces the overall efficiency of the device by lowering the $V_{oc}$. Defect states or traps introduced by certain inorganic materials, particularly metal oxides like NiOx, can further exacerbate this problem. Moreover, some inorganic HTLs, such as CuSCN, may release toxic ions that raise concerns about the environmental impact and sustainability of the solar cells (Shahinuzzaman et al. 2022). Given these limitations, CuI (Copper Iodide) is selected as the HTL in the model PSC configuration due to its favourable properties. CuI exhibits high hole mobility, which ensures efficient charge transport through the HTL and minimizes resistive losses (Matondo et al. 2021). This enhances the FF of the device and contributes to a higher overall PCE. Moreover, CuI offers excellent stability compared to organic HTLs like Spiro-OMeTAD. It is less susceptible to environmental degradation, such as moisture and temperature fluctuations, resulting in improved long-term stability for the device.

Introducing a buffer layer in PSC is essential for several reasons. Primarily, it helps in achieving optimal energy level alignment between the perovskite layer and the charge transport layers (Njema et al. 2025). This alignment is crucial as it minimizes energy loss and enhances the efficiency of charge extraction. By facilitating better energy alignment, the buffer layer ensures that electrons and holes can move more freely from the perovskite layer to the electrodes, thus improving the overall performance of the cell (Li et al. 2023a). Additionally, buffer layers play a significant role in reducing recombination losses. They act as a barrier against trap states that could otherwise cause charge carriers to recombine prematurely before reaching the electrodes. This reduction in recombination losses translates to improved charge carrier collection and, consequently, higher efficiency.

Ultimately, buffer layers protect the perovskite layer against adverse weathering conditions such heat, moisture and oxygen, and also shield the perovskite material from potential damage or degradation caused by subsequent processing steps (Ma et al. 2022). This protection is vital for maintaining the integrity and longevity of the solar cell. Buffer layers also contribute to the stability of the solar cell device. They can mitigate the effects of moisture, oxygen, and other environmental factors that might otherwise degrade the perovskite material. This enhanced stability is crucial for ensuring the long-term performance of the solar cell. Moreover, buffer layers can optimize the electrical properties of the device (Yin et al. 2017). For instance, they can improve the work function of the electrodes or enhance the overall conductivity of the cell. Some buffer materials even improve the optical properties of the device, leading to better light absorption and transmission. Generally, buffer layers are indispensable in perovskite solar cells. They address several challenges, including energy alignment, charge transfer efficiency, recombination losses, environmental stability, and optical performance. By fulfilling these roles, buffer layers significantly enhance the efficiency, stability, and overall performance of PSCs.

To align simulated device results with experimental findings using SCAPS-1D, a careful approach to parameter control and calibration is essential (Beard and Cole 2022). Starting by ensuring that the material properties used in SCAPS-1D, such as bandgap values, electron affinities, and defect densities, accurately reflect those of the experimental materials

(López-Fernández et al. 2024). Accurate input parameters are crucial for obtaining reliable simulation outcomes. Temperature effects should be incorporated into the simulation, as materials respond differently to temperature variations, which can influence carrier mobility, bandgap values, and recombination rates (Njema et al. 2024a). Making sure that simulation accounts for these temperature dependencies to better replicate experimental conditions is critical in PV technology. When modelling the absorber layer thickness, varying the thickness in the simulations to find the optimal value that corresponds with experimental results (Bimli et al. 2023). The thickness of the absorber layer can significantly impact the electrical and optoelectronic properties of the device, so it is crucial to match it closely with the experimental setup.

For the metal back contact, ensuring that the work function values and stability characteristics of materials like Pd is accurately represented in SCAPS-1D. The properties of the metal back contact play a significant role in device performance, therefore precise modelling is necessary (Matondo et al. 2021). If the experimental setup includes interface materials, it is necessary to model their effects accurately in the numerical simulator (Sajid et al. 2023a). This involves accounting for their band alignment, defect density, and how they influence charge carrier dynamics. Accurate representation of these interfaces will help to better mimic experimental conditions. Optimization techniques such as parameter sweeping can be valuable. By varying critical variables like temperature, absorber thickness, and material properties, one can understand their impact on device performance and adjusting model accordingly. Fitting simulation parameters to experimental data is a key step in refining the accuracy simulation data. It is good practice to compare simulation results with experimental data to identify discrepancies and make necessary adjustments. Cross-validating the results with other simulation tools or theoretical models can also help ensure the robustness of the findings. Finally, conducting a detailed analysis of key performance metrics such as PCE, FF, and $J_{sc.}$ Investigate any differences between simulated and experimental results to identify potential causes and refine the simulated model. By following these steps, one can enhance the alignment of SCAPS-1D simulations with experimental results, leading to more accurate and reliable predictions.

Scalable fabrication methods of perovskite solar cell architectures and cost reduction by using CuI hole transport layer material is critical for their commercialization. Accordingly, long-term stability under real-world conditions is essential for enhanced device performance. Research into CuI-based PSCs is rapidly advancing, with considerable potential for future improvements in efficiency, stability, and scalability. Whereas this study is based on theoretical predictions, it provides insights towards the fabrication of a physical solar cell architecture of robust stability, affordability, eco-friendly and modest efficiency. Therefore, it offers the promise of contributing immensely to the advancement of the photovoltaic technology.

## 2 Device simulation and numerical methodology

Solar cell capacitance device simulator SCAPS-1D is particularly effective for studying thin-film and perovskite solar cells (PSCs) due to its specialized focus, user-friendly design, and detailed physical modelling capabilities. Whereas other device simulation tools like Silvaco, COMSOL, and TCAD offer broader functionalities, SCAPS-1D's targeted approach makes it especially suited for detailed PSC research. The choice of simulation software ultimately depends on the specific requirements and objectives of the research.

SCAPS-1D established reputation within the scientific community adds to its credibility, as it has been validated in numerous studies focused on perovskite and thin-film solar cells (Hossain et al. 2023a). This validation contributes to the confidence researchers can place in its results. Towards this end, we have used SCAPS-1D device simulator under standard illumination power of one sun of 1000 W/m$^2$ under 1.5 AM (Zollinger and Cole Whitehouse 2022) to mimic real world conditions in device investigation. Actual experimental and simulation data from published works have been used while optimizing the thickness of the absorber in order to validate the data reported (Chowdhury et al. 2020). SCAPS-1D device simulator solves the continuity, Poisson and carrier transport equations for excitons.

The Poisson expression is given by Eq. 1:

$$\frac{dy}{dx}\left(\varepsilon_0\varepsilon\frac{\partial\varphi}{\partial x}\right) = -q\left(p - n + N_D^+ + -N_A^- - + \frac{\rho def}{q}\right) \tag{1}$$

Here, $\varphi$ denotes the electrostatic potential, $n$ and $p$ are the free exciton carriers per unit volume, $\varepsilon_0$ and ε denote the permittivity of vacuum and semiconductor, $N_D^+$ and $N_A^-$ are the density of ionized donors and acceptors whereas *def* is the defect charge density.

The continuity equations for electron and hole can be expressed as Eqs. 2–5.

$$\frac{\partial n}{\partial t} = -\frac{\partial}{\partial \mathrm{x}}\mathrm{Jn} - \mathrm{Un} + \mathrm{G} \tag{2}$$

$$\frac{\partial p}{\partial t} = -\frac{\partial}{\partial x}Jp - Up + G \tag{3}$$

$$\mathrm{Jn} = -\frac{\nu\mathrm{nn}}{\mathfrak{q}}\frac{\partial \mathrm{Efp}}{\partial \mathrm{x}} \tag{4}$$

$$\mathrm{Jp} = +\frac{\mathrm{upP}}{\mathrm{q}}\frac{\partial \mathrm{Efp}}{\partial \mathrm{x}} \tag{5}$$

Here, *Jn* and *Jp* are the electron and hole current density, respectively. *G* and *U* are the generation and recombination rates of charge carriers whereas *Efn* and *Efp* are the electron and hole Fermi levels, respectively. The basic input parameters of the model solar cell structure are presented in Table 1.

## 3 Results and discussion

The p–n junction solar cell architecture is composed of a number of stack levels as can be seen from the proposed cell architecture (Fig. 1a). FTO serves as an electrode on the top layer of the solar cell, allowing photons to penetrate the structure and transport photogenerated electrons to the terminals of external. The second layer is an electron transport layer, which serves as a hole-blocking layer by limiting charge recombination and is utilized to collect photogenerated electrons. The perovskite absorber layer, which is the third layer, is utilized to absorb light and generate excitons. When subjected to an electric field, these photo-generated free charge carriers scatter and drift across the terminals. A hole transport layer, which is the fourth layer, improves device stability. The fifth layer the palladium back contact. In essence, the functionality of each component within the perovskite solar cell

**Table 1** Input parameters for the proposed cell architecture

| | $CH_3NH_3SnI_3$ (Du et al. 2016) | CuI (Al-Mousoi et al. 2023) | $TiO_2$ (Tiwari and Belwal 2024) | FTO (Tiwari and Belwal 2024; Deswal et al. 2024) |
|---|---|---|---|---|
| Thickness (nm) | 1000 | 90 | 100 | 50 |
| Bandgap (eV) | 1.3 | 1.290 | 3.2 | 3.5 |
| Electron affinity (Ev) | 4.170 | 4.200 | 4.0 | 4.0 |
| Dielectric permittivity (relative) | 6.500 | 3.000 | 9.0 | 9.0 |
| CB effective density of states | $1\times10^{18}$ | $2.2\times10^{18}$ | $1\times10^{19}$ | $9.2\times10^{18}$ |
| VB effective density of states | $1\times10^{19}$ | $1.8\times10^{19}$ | $1\times10^{19}$ | $1.\times10^{19}$ |
| Electron mobility ($cm^2$/Vs) | 1.6 | $1.5\times10^{2}$ | 2.0 | $2\times10^{1}$ |
| Hole mobility ($cm^2$/Vs) | 1.6 | $1\times10^{2}$ | 1.0 | $1\times10^{1}$ |
| Shallow uniform donor density | 0 | 0 | $1\times10^{16}$ | $1\times10^{19}$ |
| Shallow uniform acceptor density | $1\times10^{15}$ | $1\times10^{15}$ | 0 | 0 |
| Defect density | $1\times10^{11}$ | $1\times10^{17}$ | 0 | 0 |

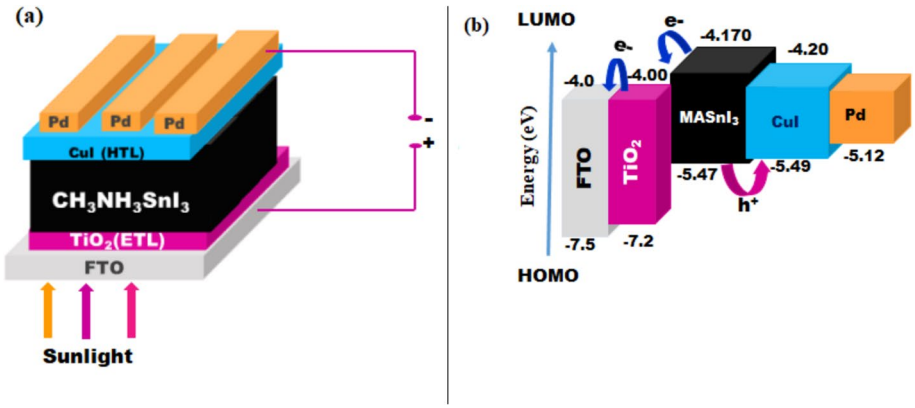



**Fig. 1** The solar cell architecture **a** and **b** the corresponding band alignment

setup is crucial for optimal charge extraction and separation. Through precise alignment of energy levels (Fig. 1b) and careful engineering of materials, the device ensures efficient movement of excitons from the absorber layer through electron and hole transport layers to the external circuit (Fig. 1a). This orchestrated process minimizes recombination losses, enhancing overall solar cell efficiency. By understanding the intricate energy dynamics and functions of each layer, scientists aim to further refine and optimize perovskite solar cell technology for practical applications in renewable energy generation.

The LUMO and the HOMO of the perovskite are −4.170 and −5.47 eV, respectively, and allows for efficient light absorption and charge generation. When photons from sunlight strike the absorber, excitons (electron–hole pairs) are formed and subsequently dissociate into free charge carriers. Electrons move toward the ETL, while holes migrate toward the HTL (CuI). The slight energy difference between the LUMO of the perovskite and the

conduction band of $TiO_2$ facilitates this directional flow of electrons toward the FTO. CuI, has a LUMO level of −5.49 eV and a HOMO level of −5.12 eV. It plays a vital role in ensuring that holes generated in the perovskite are efficiently transferred to the back contact. The HOMO of CuI aligns closely with the HOMO of the perovskite, which ensures minimal energy loss during hole transfer. This smooth transport of holes through CuI leads to their collection at the back contact. Palladium has a high work function of 5.12 eV which makes it ideal for collecting holes from the CuI HTL layer. Its stable and conductive nature ensures that holes can be extracted with minimal resistance, further enhancing the solar cell's performance. Overall, the energy band diagram demonstrates the importance of precise band alignment between the various layers to facilitate efficient charge transport.

### 3.1 Quantum efficiency and J–V analysis

Quantum efficiency (QE) is a measure of the ability of a solar cell to convert incident photons into electrical current. QE is typically measured as a function of wavelength or energy, and can provide import insights into the performance and efficiency of a solar cell device. In general, the QE of a solar cell increases with increase in the thickness of the device. This is because a thicker cell can absorb more light and therefore convert more photons into electrical current (Jiang et al. 2023; Tang et al. 2023); however, there is a limit to this effect, beyond which the quantum efficiency starts to decrease. A reduction of overall QE is caused by reflection and low diffusion length, reduced surface recombination and reduced absorption at long wavelengths. No light is absorbed below the band gap so the QE approaches low values at long wavelengths as it can be observed in Fig. 2. The exact relationship between quantum efficiency and thickness can depend on the specific design and materials used in solar cells. For instance, if the thickness of the cell is too large, the

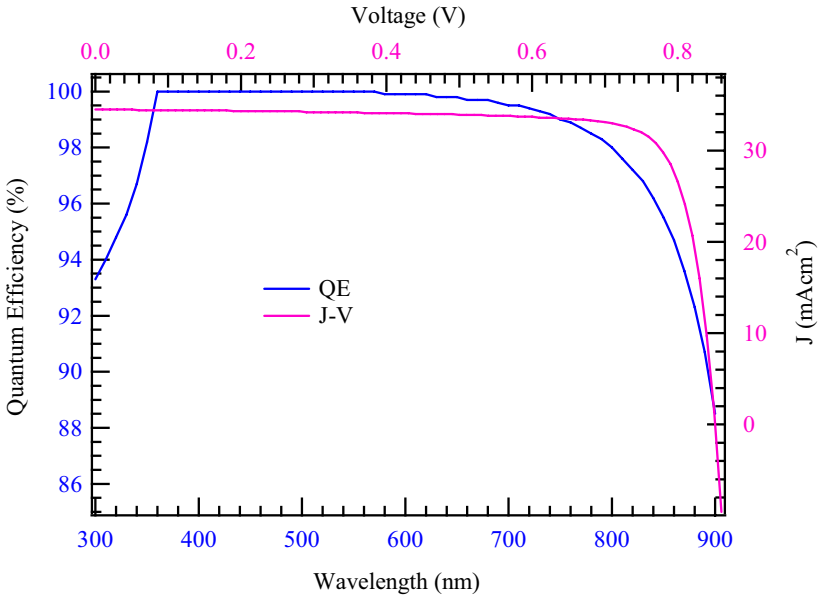



**Fig. 2** QE and J-V characteristics of the model cell structure

extra material may actually reduce the efficiency by scattering the photons and reducing their ability to generate current (Zhou et al. 2023). If the thickness of the cell is not uniform throughout, there may be variations in the quantum efficiency across the surface of the cell. These variations can be caused by changes in the absorption properties of different areas of the cell, or by the presence of defects or impurities (Hossain et al. 2023b).

From Fig. 2 wavelength (nm) delineates the spectrum of light wavelengths to which the solar cell was exposed, ranging from 300 to 900 nm, encompassing the ultraviolet (UV), visible, and near-infrared (NIR) regions of the electromagnetic spectrum. Voltage (V) axis, the graph illustrates the voltage output of the solar cell corresponding to different wavelengths of light. Voltage gradually rises from 0 V to approximately 0.6 V at $J_{sc}$ value of 33 $mAcm^2$. The QE showcases the solar cell efficiency in transforming photon energy into electrical energy. QE rises from around 93% at 300 nm to its peak of about 100% at 350 nm, and continues to remain at 100% up to about 650 nm before steadily to 84% at 900 nm. The $J_{sc}$ denotes the short-circuit current density, representing the maximum current output under illumination, typically measured in $mA/cm^2$. The J–V is indicative of the current density–voltage (J–V) curve, a crucial tool for evaluating solar cell performance by illustrating the relationship between current density (J) and voltage (V). Overall, Fig. 2 suggests optimal efficiency of the solar cell in converting light occurs around the 350 nm wavelength, situated in the blue region of the visible spectrum. Notably, the cell exhibits responsiveness across a broad spectrum, spanning from UV to the near infrared region.

The J–V characteristics summarize their effectiveness and operation of perovskite solar cells. These cells usually display high $V_{oc}$ and significant $J_{sc}$, leading to notable power conversion efficiencies. Yet, issues such as hysteresis, stability, and consistency remain unresolved. Ongoing work in material design engineering, device structure, and interface enhancement seeks to enhance J–V curves, with the goal of achieving enduring advancements in efficiency, stability, and scalability. These improvements are essential for the widespread adoption of perovskite solar cells in mainstream photovoltaic applications. Moreover, factors such as cell design and material quality also impact solar cell efficiency.

### 3.1.1 Quantum efficiency analysis for different buffer layers

Buffer layers significantly influence the QE of a solar cell structure by enhancing charge collection and reducing energy losses at interfaces (Karthick et al. 2021). In PSCs, buffer layers such as CdS, ZnSe or $TiO_2$ are positioned between the perovskite absorber and transport layers to improve charge extraction and minimize interfacial recombination (Njema et al. 2024a). These layers create energy barriers that facilitate efficient electron or hole transport, thus increasing the photogenerated current and QE. The effectiveness of the buffer layer depends on its material and thickness; a well-chosen buffer layer can boost light absorption and lower surface recombination losses, leading to higher QE and PCE (Thakur et al. 2022). Conversely, an inappropriate buffer layer can introduce additional recombination sites or hinder charge transport, negatively affecting both QE and overall device performance.

The QE characteristics of the solar cell structure, as depicted in Fig. 3, demonstrate how different buffer layers – CdS, ZnSe, and ZnS impact the cell's ability to convert incident photons into electrical current across various wavelengths (Khaleda et al. 2021). The CdS and ZnSe buffer layers maintain high QE in the entire wavelength range. This indicates that the buffers facilitate effective electron transport and minimizes recombination losses, contributing to the cell's remarkable performance (Bishnoi et al. 2022). However, a

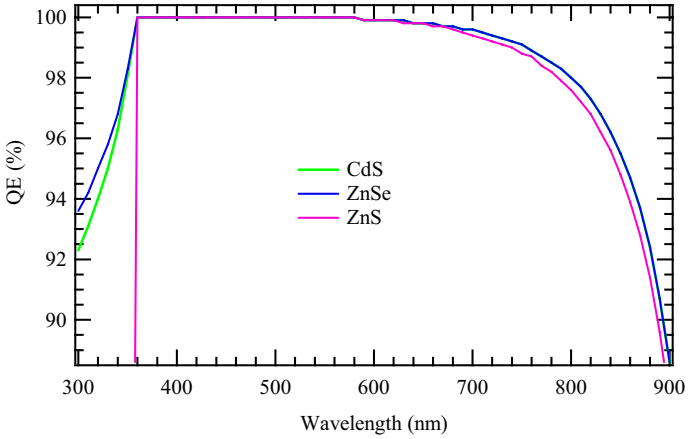



**Fig. 3** The impact of buffer layers on the QE characteristics of the cell structure

considerable drop in QE is observed in the blue region of the spectrum, which is a known limitation of CdS and ZnSe due to their bandgap of approximately 2.42 and 2.7 eV, respectively (Mohamed et al. 2020). This drop signifies absorption losses at shorter wavelengths, reducing its effectiveness in converting high-energy photons to electrical current. ZnSe, nevertheless, offers potential advantages in terms of greater stability and lower toxicity compared to CdS, making it an attractive alternative in semiconductor applications. On the other hand, ZnS buffer layer, represented by the magenta curve, exhibits the lowest overall QE across most of the spectrum, particularly in the blue region. This is largely due to ZnS's wide bandgap of around 3.54 eV (Phogat et al. 2022), which limits its ability to absorb visible light effectively. Consequently, the QE is significantly lower, reducing the overall efficiency of the cell. Despite this, ZnS is valued for its non-toxic nature and excellent chemical stability, making it an environmentally friendly choice even though it compromises on efficiency. The graph displays the performance of CdS, ZnSe, and ZnS as buffer layers in a solar cell. CdS demonstrates a marginally higher QE, reaching up to 90% between 350 and 550 nm. Both ZnSe and ZnS show similar trends, achieving around 85% efficiency, particularly in the visible light spectrum (400–700 nm). All three materials experience a sharp decline in QE beyond 700 nm.

### 3.2 The impact of absorber thickness on cell performance

Thickness is an important factor in determining the performance of PSCs (Los Santos et al. 2020; Katunge et al. 2023). Its impact has been studied theoretically in this work. The active layer thickness was varied from 250 to 3000 nm typically in steps of 200 nm. As can be noted from Fig. 4a, $J_{sc}$ increases significantly with the increasing thickness of the absorber until it reaches a maximum value of 34.90 $mAcm^{-2}$ between 2300 and 3000 nm.

The thickness of a solar cell can be optimized to maximize its $J_{sc}$. A thicker cell will absorb lighter and produce more electron–hole pairs which can increase $J_{sc}$. However, if the cell is too thick, some of the light will be absorbed before it reaches the active region of the cell, thus decreasing the value of $J_{sc}$. On the other side, the $V_{oc}$ depicted in Fig. 4b shows a notable rise as the absorber thickness increases, peaking at 0.88 V at about 1500 nm. From Fig. 4c the PCE increased from 12.63% to an optimal value of 23.04% at a thickness

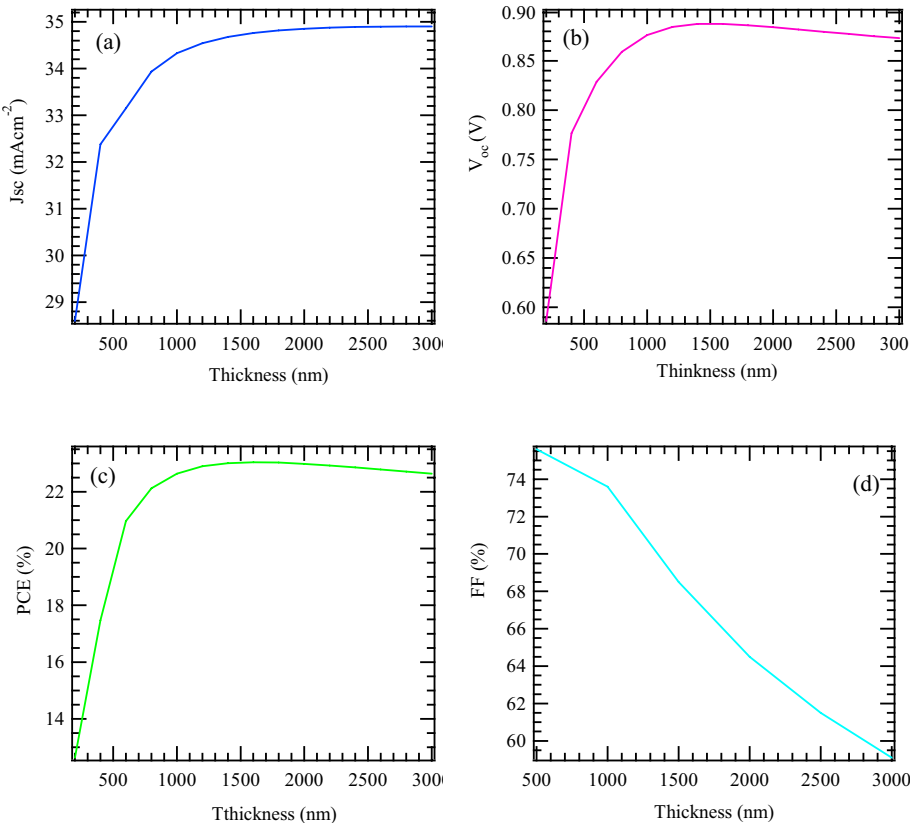



**Fig. 4** The effect of absorber thickness on electrical performance of the model cell

of 1500 nm. Afterwards it decreased to a value of 22.64% at 3000 nm thickness. This can be attributed to the fact that with increase in the thickness of the perovskite layer, more energy is absorbed and this may lead to increased recombination of the current in the solar cell. The FF according to Fig. 4d decreases in a steady version from approximately 76% at 500 nm thickness to approximately 59% at 3000 nm.

### 3.3 Influence of temperature on the cell performance

The operating temperature has a significant impact on the solar cell's performance (Njema and Kibet 2023a). Even though a solar cell's normal operating temperature is 300 K (Bouazizi et al. 2022), its temperature will rise dramatically while it is in use. The current–voltage (JV) curves for PV cells are strongly impacted by temperature fluctuation (Torres et al. 2018). Generally, solar devices are intended to function under room temperature conditions; nonetheless, the operational performance of a solar cell structure decreases with increase in temperature (Li et al. 2023b; Gad et al. 2023). Therefore, it is crucial to examine the influence of the operating temperature on the PV performance when designing a solar cell structure. In this study, the operating temperature on the proposed lead-free $CH_3NH_3SnI_3$-based PSC is varied from 260 to 400 K. $V_{oc}$ falls steadily in the entire

temperature range as depicted by Fig. 4a. $J_{sc}$ decreases gradually with increase in temperature according to Fig. 4b. The saturation current, which may vary by orders of magnitude, affects $J_{sc}$. Equation 6 demonstrates how $V_{oc}$ is dependent on both the light-generated current and the solar cell's saturation current.

$$\mathrm{Voc} = \frac{K_B T}{q} \mathrm{In}\left(\frac{J_{ph}}{Jo} + 1\right) \approx \frac{K_B T}{q} \mathrm{In}\left(\frac{J_{ph}}{Jo}\right) \tag{6}$$

The $V_{oc}$ is sharply reduced with increase in the operating temperature, and this is because the reverse saturation current which increases with increasing temperature (Bhari et al. 2023; Khatoon et al. 2023). In a material, as the thermal energy increases, the energy of electrons increases, and therefore highly energetic electrons have a high potential to recombine with holes. Ultimately, increased exciton recombination results in decreased $V_{oc}$ as observed in Fig. 5a. Higher or considerably low temperatures can affect efficiency; therefore, moderate temperatures may be ideal for solar panel performance. A solar panel's output can drop by between 0.3 and 0.5% for every degree rise over 25 °C, which has a significant impact on overall energy generation (Williams et al. 2023).

Typically, as temperature rises, $V_{oc}$ decreases. This occurs because the semiconductor material's bandgap decreases with temperature. When a photon is absorbed by the semiconductor, it can elevate an electron from the valence to the conduction band, and

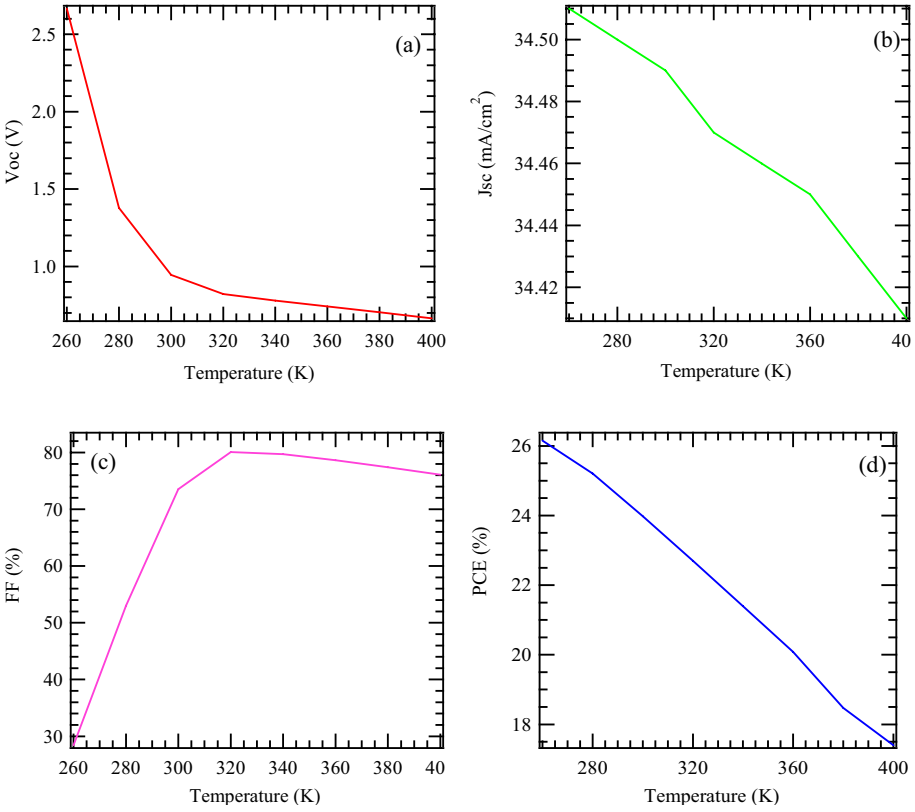



**Fig. 5** Effect of temperature on electrical output

the disparity in energy between these bands dictates the solar cell's voltage output. In Fig. 5b, a direct negative correlation between temperature and $J_{sc}$ occurs, although the change in $J_{sc}$ as temperature increases is small because at 260 K, $J_{sc}$ is 34.51 $mAcm^{-2}$ while at 500 K, $J_{sc}$ is 34.27 $mAcm^{-2}$. This implies that the model cell device has a good temperature tolerance. The small change may be ascribed to the fact that as the temperature increases; thermal vibrations occur which may increase the resistance of the material slightly; however, this phenomenon is material dependent.

Meanwhile in Fig. 5c, the FF indicates the efficiency of converting light into electricity in the model solar cell. It is observed that FF increases as temperature rises but later declines after peaking at about 80% at 320 K. This decline signifies that the efficiency of the solar cell diminishes as it gets hotter. This phenomenon occurs because solar cells tend to be less effective at converting light into electricity under higher temperature conditions. Initially, the FF of the cell increases with temperature according to Fig. 5c. At 260 K, the FF of the proposed solar cell is approximately 44%. As the temperature increases, the FF starts to decrease, and this is attributed to the fact that temperature affects the electron and hole mobility. Generally, increased temperature affects parameters such as carrier mobility; exciton concentration and the bandgap of the material resulting to increased cell resistance and thus, the electrical output of the cell structure are observed to decrease at elevated temperatures.

Figure 5d shows a clear negative relationship between temperature and PCE. As the temperature increases, the PCE declines. The PCE of the device decreases from 25% at 260 K to 12.73% at 400 K as can be deduced from Fig. 4d. Decrease in PCE temperature dependent, although this can be influenced bandgap, charge mobility, and exciton concentration (Ali et al. 2023). This is because higher temperatures increase the thermal energy of the electrons in the PV cell, causing excitons to jump across the energy gap and be lost as heat instead of contributing to the electrical output (Lin and Wang 2022). Moreover, the performance of the materials used in the PV cell can degrade at higher temperatures, further decreasing the PCE.

In PSCs, the $V_{oc}$ plays a crucial role in determining device efficiency, and its relationship with temperature is key to understanding the overall performance and stability of the cells. Generally, $V_{oc}$ decreases as temperature increases. This behavior is due to the temperature-sensitive nature of the semiconductor materials used in PSCs (Njema and Kibet 2023b). Specifically, as temperature rises, the reverse saturation current density increases exponentially because of enhanced carrier recombination, which leads to a reduction in $V_{oc}$ (Tress et al. 2018). This reduction is typically linear and characterized by a negative temperature coefficient, with typical values ranging from $-2.2$ to $-2.5$ mV/°C, though this can vary depending on the specific materials and cell architecture (Mueller et al. 2003).

The relationship between $V_{oc}$ and temperature is further connected to the activation energy (Ea) of recombination processes within the solar cell. The activation energy is an important parameter that reflects the energy barrier for recombination, particularly at interfaces and within the bulk of the perovskite layer (Shao and Loi 2020). In highly efficient PSCs, the activation energy for recombination is usually close to the bandgap energy of the perovskite material (Chen and Park 2019). If the activation energy is significantly lower than the bandgap, it indicates that recombination predominantly occurs through trap states or at interfaces, rather than within the bulk material (Shukla et al. 2021). This type of non-radiative recombination leads to lower $V_{oc}$ and consequently reduces the overall efficiency of the solar cell (Njema et al. 2024b). In PSCs, the $V_{oc}$ is closely linked to the activation

energy of the recombination processes within the device. This relationship is crucial for understanding the efficiency and identifying loss mechanisms in solar cell devices.

The activation energy, meanwhile, is the energy barrier that must be overcome in order to minimize recombination, and is typically associated with the recombination processes happening within the device, particularly at the hetero-interfaces or within the bulk materials. This implies that the recombination is primarily radiative, which is less harmful to $V_{oc}$ since the energy lost in these processes can be partially recovered in the form of emitted light (Guo et al. 2022). However, if the activation energy is significantly lower than the bandgap, it indicates the presence of non-radiative recombination pathways, such as defects or trap states either within the perovskite layer or at the interfaces between different materials (Luo et al. 2020). Non-radiative recombination is particularly detrimental to $V_{oc}$ because it leads to energy losses that are not converted into electrical output (Tress et al. 2015). These losses reduce the quasi-Fermi level splitting, thereby lowering the $V_{oc}$. The extent of this reduction is directly tied to the activation energy: a lower activation energy means more pronounced non-radiative recombination, resulting in significant loss in $V_{oc}$ (Guo et al. 2022).

Understanding this relationship underscores the importance of optimizing both the material properties and the interfaces within PSCs to maximize $V_{oc}$. Reducing the density of defect states, which facilitate radiative recombination, is key to improving $V_{oc}$ (He et al. 2022). This can be achieved through better material synthesis, effective passivation of surfaces and interfaces, and the use of high-quality transport layers that reduce defect formation. Therefore, the relationship between $V_{oc}$ and temperature, as well as the role of activation energy in recombination processes, is essential for improving the performance and stability of PSCs.

### 3.4 Effect of changing back contact material on electrical parameters

Back contact materials for use in PSCs have received a lot of attention since they are important in determining the electrical and optoelectronic output of the cell. However, lack of thorough understanding of their operation from device physics standpoint has prevented them from reaching their full potential**.** In this study palladium is used as aback contact because it is a promising candidate since it has a relatively high work function, which means it can easily extract electrons from the semiconductor material and enhance the built-in electric field in the depletion region of the solar cell (Arif et al. 2024; Thomas et al. 2022). This can lead to higher open circuit voltage and overall efficiency of the solar cell. Moreover, palladium has good stability under ambient conditions and in contact with the semiconductor material. This stability can improve the long-term performance and reliability of the solar cell. Also, palladium has a low resistivity, which can reduce the series resistance of the solar cell and improve the FF, and hence the efficiency of the cell. Palladium has a low reflectivity in the visible and near-infrared regions of the spectrum, which can enhance the absorption of light in the semiconductor material and improve the $J_{sc}$ of the solar cell. Palladium is compatible with various thin-film technologies such as copper indium gallium selenide (CIGS), cadmium telluride (CdTe), and organic photovoltaic (OPV) cells, making it a versatile back contact material option for different types of solar cells (Soonmin et al. 2023).

The back contact plays an important role in determining the efficiency of a solar cell. The FF is affected by several factors, including the efficiency of the cell's energy conversion, the resistance of the cell, and the recombination rate of the carriers. Changing

the back contact of a solar cell can affect the FF in several ways. For example, changing the back contact material or structure can affect the carrier recombination rate and the cell's resistance. This, in turn, can affect the FF. In general; changing the back contact to a more efficient material or structure can improve the cell's efficiency and increase the FF as noted in Table 2. Nevertheless, if the change in the back contact results in increased carrier recombination or increased resistance, the FF can decrease. On the other hand, changing the back contact of a solar cell can affect the PCE in several ways. For example, changing the back contact material or structure can affect the carrier recombination rate, carrier collection efficiency, and the resistance of the cell. If the new back contact material or structure reduces carrier recombination, improves carrier collection efficiency, and reduces resistance, the PCE can increase.

The $J_{sc}$ is affected by several factors, including the absorption of light in the cell, the carrier lifetime, and the carrier collection efficiency. Changing the back contact of a solar cell can affect the $J_{sc}$ in several ways. For example, changing the back contact material or structure can affect the absorption of light in the cell and the carrier collection efficiency. If the back contact material or structure is more reflective or absorbs less light, the $J_{sc}$ can decrease. On the other hand, if the new back contact material or structure improves the carrier collection efficiency, the $J_{sc}$ can increase. This observation can be clearly noted in Table 2. Although Pt a robust PCE of 27.14%, its FF of 33.29 is significantly compromised and therefore may be a good back contact for the model cell structure. On the hand, gold (Au) gives remarkable performance metrics—PCE of 23.95% but with slightly lower FF as compared to that of Pd. Notably, Au is very expensive, and therefore, Pd is preferred owing to its affordability. Another back contact of interest is Cobalt with a better FF (79.70%) but boasts a slightly lower PCE and $V_{oc}$ in comparison to Pd.

Therefore, the effect of changing the back contact on the $J_{sc}$ of a solar cell depends on several factors, including the material and structure of the back contact, the absorption of light in the cell, and the carrier lifetime and collection efficiency. Changing the back contact material or structure can affect the bandgap of the semiconductor material and the temperature of the cell. If the new back contact material or structure has a different work function or thermal conductivity, the $V_{oc}$ of the material changes correspondingly. The $V_{oc}$ depends on the material and structure of the back contact, the bandgap of the

**Table 2** Effect of changing metal back contact on $V_{oc}$, FF, $J_{sc}$ and PC

| | $V_{oc}$ (V) | Jsc (mA/cm$^2$) | FF (%) | PCE (%) | Work function (eV) |
|---|---|---|---|---|---|
| Palladium | 0.8762 | 34.33 | 75.27 | 22.64 | 5.12 |
| Nickel | 0.9497 | 34.49 | 73.10 | 23.95 | 5.15 |
| Platinum | 2.3313 | 34.97 | 33.29 | 27.14 | 5.65 |
| Gold | 0.9214 | 34.45 | 74.62 | 23.69 | 5.10 |
| Molybdenum | 0.4237 | 33.88 | 71.58 | 10.28 | 4.6 |
| Cobalt | 0.8178 | 34.36 | 79.70 | 22.40 | 5.0 |
| Iron | 0.3238 | 33.66 | 67.06 | 7.31 | 4.5 |
| Silver | 0.1168 | 32.70 | 46.68 | 1.78 | 4.26 |
| Copper | 0.5237 | 34.04 | 74.69 | 13.32 | 4.7 |

semiconductor material, and the temperature and concentration of dopants in the material. The influence of back contact on the electrical outcomes of the model cell is reported in Table 2.

From Table 2, $V_{oc}$ for the Pt-based structure is remarkably high at 2.3313 V, a value that far surpasses those observed for other metals, which range from 0.1168 V for Ag to 0.9497 V for Ni. Such an elevated $V_{oc}$ is atypical for PSCs, suggesting either an exceptional alignment of energy levels between the perovskite absorber and the Pt contact or the presence of a possible anomaly in the simulation process. One plausible explanation for this unusually high $V_{oc}$ lies in the relatively high work function of Pt (5.65 eV), which could theoretically lead to a larger built-in potential at the interface with the perovskite material, thereby increasing $V_{oc}$; however, the magnitude of this $V_{oc}$ is matched with a low FF of 33.29%. Typically, a low FF indicates issues with charge extraction or increased recombination losses, which would not ordinarily be associated with a high $V_{oc}$. Despite the high $V_{oc}$ and relatively an impressive PCE of 27.14%, the low FF suggests significant inefficiencies, likely due to poor interface quality or improper energy alignment. This results in suboptimal charge transport. In contrast, Pd and Ni exhibit $V_{oc}$ values of 0.8762 and 0.9497 V, respectively, which are within the expected range for PSCs. These metals also show relatively high FF values, indicating effective charge extraction and low recombination losses, contributing to decent PCE values of 22.64% for Pd and 23.95% for Ni. Au, with a $V_{oc}$ of 0.9214 V, achieves a similar FF to Pd but shows a slightly lower PCE, possibly due to its slightly lower work function of 5.10 eV, which may affect the built-in potential and thus the $V_{oc}$. Ag presents the lowest $V_{oc}$ at 0.1168 V, indicating poor energy level alignment with the perovskite absorber, leading to the lowest PCE of 1.78%.

In conclusion, the extraordinarily high $V_{oc}$ for the Pt-based structure raises concerns about the accuracy of the simulation or suggests the presence of unusual physical phenomena that merit further investigation. The combination of a low FF and a high $V_{oc}$ indicates that, although a strong built-in potential may exist, significant losses in charge collection are likely occurring due to recombination at the interfaces. To confirm these results and fully understand the underlying mechanisms, further scrutiny of the simulation parameters and experimental validation is essential. The performance of solar cells, particularly their PCE, is influenced by various factors, including the work function of the metal used as the back contact. The work function directly affects key parameters like the $V_{oc}$, $J_{sc}$, and FF. In this context, the Pd device shows a slightly lower PCE of 22.64% compared to the Gold (Au) device, which has a PCE of 23.69%. Although the work function of Pd (5.12 eV) is very close to that of Au (5.10 eV), the Pd device exhibits a lower Voc of 0.8762 V, whereas the Au device achieves a Voc of 0.9214 V. The work function plays a crucial role in determining the energy level alignment at the metal/semiconductor interface, which influences the efficiency of charge extraction and the rate of recombination within the device. When the difference in work function between the back contact and the semiconductor is minimized, energy level alignment improves, reducing the barrier for charge extraction and limiting recombination losses.

However, despite its slightly higher work function, the Pd device produces a lower $V_{oc}$ compared to the Au device. This discrepancy can be attributed to several factors. One possibility is the presence of a higher density of interface states or defects at the Pd/semiconductor interface, which could act as recombination centers, thereby reducing the $V_{oc}$ and overall PCE. Moreover, the lower $V_{oc}$ suggests that the Pd device may suffer from higher recombination losses, either at the interface or within the bulk of the absorber. These losses might stem from less optimal energy level alignment or poorer passivation at the interface

compared to the Au device. While the Pd device exhibits a slightly higher FF (75.27%) than the Au device (74.62%), indicating potentially better charge extraction efficiency or lower series resistance, the reduced $V_{oc}$ possible increase in recombination overshadows this advantage, leading to a lower overall PCE. This highlights the delicate balance between work function and interface properties in achieving optimal solar cell performance.

### 3.5 Impact of perovskite thicknesses on electrical characteristics the cell

More sun exposure is correlated with thicker absorber thickness, which increases light absorption. More surplus charge carriers, (holes and electrons) will eventually be created and transported to the appropriate electrodes in order to produce electricity. The photoactive layer thickness has a major impact on the device's PCE in a solar cell. The absorption rate of an ultrathin layer is reduced, resulting in a lower PCE for the cell. It is evident that higher PCE is mainly associated with higher thickness of the absorber layer due to less recombination at the back surface. J-V characteristics of the solar cell greatly rely on the thickness of the perovskite material (Roy and Khare 2022). Figure 6a shows that the optimal thickness of the photoactive layer for the proposed solar cell is 2500 nm. Below this thickness, the solar cell performance decreases drastically.

The QE of a solar cell shows the capacity of a solar cell to collect excitons from incident photons of a certain wavelength. Figure 6b depicts that the photon absorption for the proposed solar cell increases with increase in the thickness of the photoactive layer. Clearly, a thickness of 2500 nm is sufficient to obtain optimal performance of the proposed solar cell. At lower thicknesses of the absorber, the QE is compromised. For instance, at 500 nm, the QE is severely affected which potentially decreases the overall performance of the cell.

### 3.6 The absorption characteristics of the solar cell components

The efficiency of a photovoltaic cell can be defined as the ratio of electrical power output to light-induced energy, signifying the cell's ability to convert energy between different forms. The parameters of the available light, including its wavelength and intensity, as well

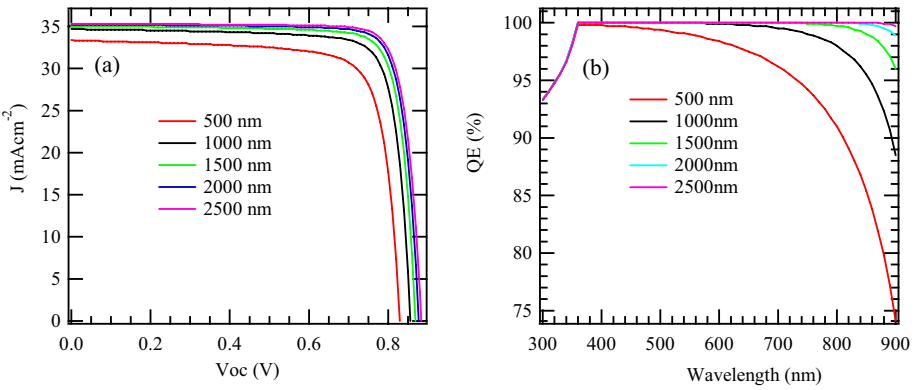



**Fig. 6** J–V characteristics. **a** and **b** Quantum efficiencies simulated at different absorber thicknesses

as the cell's various performance attributes, determine the quantity of energy generated by photovoltaic cells.

The band-gap, or the range of light wavelengths that a material can absorb and transform to electrical energy, is a crucial characteristic of photovoltaic semiconductors. PV cells are capable of efficiently utilizing all available energy if the semiconductor's band-gap aligns with the wavelengths of light shining on them. From Fig. 7 both the device performance and light absorption are improved in the longer wavelength region by the solar cells. Absorption is higher in the shorter wavelength region, but device performance suffers as a result (Jiang et al. 2023). This behaviour is similar for the absorption of Perovskite HTM and ETL, respectfully. Figure 7 clearly shows that both $CH_3NH_3SnI_3$ and CuI have the highest absorption coefficient $2.0 \approx 2.0 \times 10^5 cm^{-1}$ which makes them excellent light harvesting materials. However, it is worth noting that all the three absorption coefficients of the three layers decreases with increase in wavelength and this phenomenon compromises exciton promotion from the valence band to the conduction band.

### 3.7 Mott–Schottky capacitance analysis

Mott-Schottky analysis (MSA) emerges as an important technique in investigating the electrical properties of semiconductor materials (Rogé et al. 2024), extending its applicability even to unconventional cases like methylammonium tin iodide ($CH_3NH_3I$) within perovskite solar cells. While $MASnI_3$ iodide is not a traditional semiconductor, its role in perovskite solar cells makes it significant. Through MSA, researchers can determine the carrier type (n-type or p-type) and doping concentration of $MASnI_3$, providing insights essential for optimizing the performance of perovskite solar cells. This analysis probes the understanding of doping concentration in perovskites, particularly those employing mixed cations like methylammonium. By examining carrier density, MSA sheds light on crucial

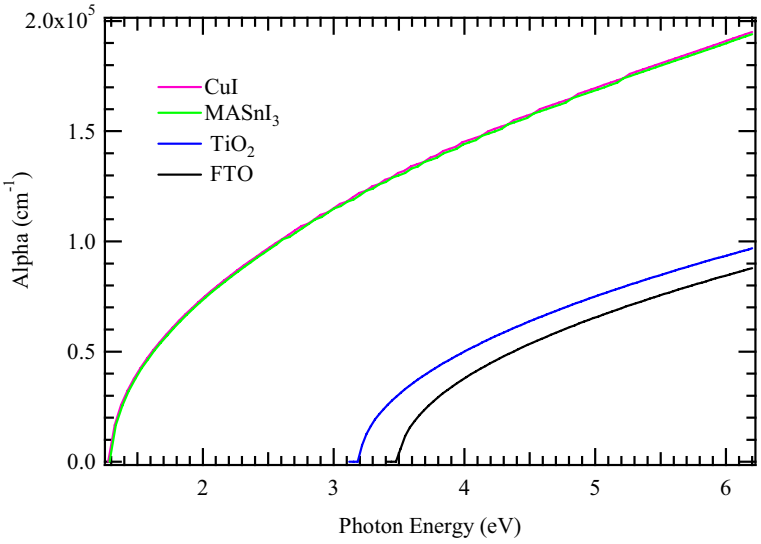



**Fig. 7** The SCAPS-1D absorption model of the proposed cell architecture

parameters affecting the device's performance, contributing to advancements in solar cell technology.

Moreover, MSA facilitates the investigation of interface properties crucial for perovskite solar cells' efficiency (Chen et al. 2023). Analysing devices with a perovskite layer containing $MASnI_3$ and a Schottky contact, forming a rectifying junction is critical in interface engineering. Through capacitance–voltage (C–V) measurements, MSA provides valuable information about the depletion region's width and carrier density at the interface (Al-Qassem et al. 2023). This insight aids in optimizing the design of perovskite solar cells, enhancing their overall performance and stability. Thus, Mott–Schottky analysis serves as a powerful tool in unravelling the electrical behaviour of $MASnI_3$ within the context of perovskite solar cells, offering crucial data necessary for advancing solar cell technology towards more efficient and sustainable energy solutions.

In Fig. 8, the accumulation of voltage over time reflects the overall electrical charge transferred within a circuit. This accumulation, measured in coulombs (C), correlates directly with the energy expended in the circuit, depicting the effort exerted by the voltage to facilitate charge movement. It can be noted from Fig. 8 that the built-in voltage of the cell is + 0.63 eV which is consistent with the value obtained from the band alignment diagram of + 0.61 eV (cf. Figure 10). This parameter is very important in determining current flow; it denotes the voltage in the depletion layer. The built-in voltage can be determined from Eq. 7.

$$V_{built-in} = \frac{K_B T}{q} In\left(\frac{N_D N_A}{n_i^2}\right) \tag{7}$$

Here, $V_{(built\text{-}in)}$ = built-in potential, $N_D$ is the n-type donor atoms, $N_A$ is the p-type acceptor atoms, $\frac{K_B T}{q}$ is the thermal voltag, n is the cncentrationof electrons, q is charge wheres T is absolute temperature.

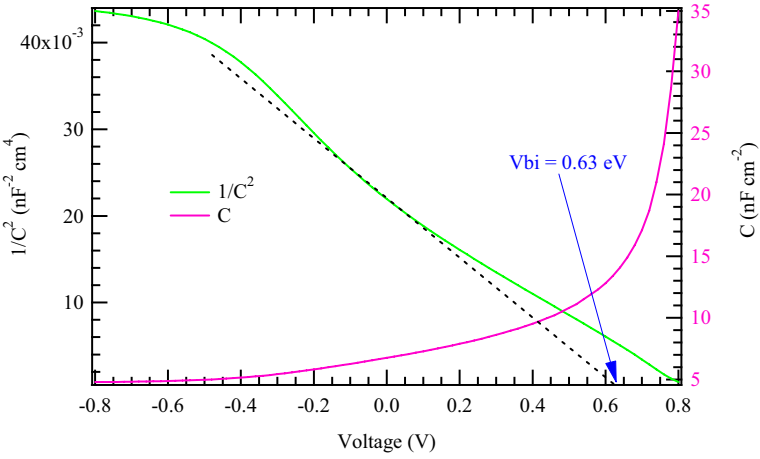



**Fig. 8** Mott-Schottky capacitance analysis

### 3.8 Nyquist plot analysis

Continuously optimizing the performance of $MASnI_3$ as a lead-free alternative for PSCs remains a focal point for researchers in the photovoltaic community (Zhao et al. 2023). Employing Nyquist plots proves instrumental in characterizing the electrochemical impedance of such devices. In a Nyquist plot specific to $MASnI_3$ within a perovskite solar cell, the following observations are noteworthy. Semicircles typically appear, representing resistances and capacitances across various interfaces within the cell. One or two semicircles may manifest at distinct frequencies. The high-frequency (HF) semicircle reflects the charge transfer process at the hole transporting layer (HTL)/perovskite interface (Yekani et al. 2024). Its diameter correlates with the charge transfer resistance at this particular junction. In certain $MASnI_3$ devices, an additional semicircle may appear, potentially attributed to grain boundaries within the perovskite layer or the interface between the perovskite and the ETL (Liang et al. 2023). At lower frequencies, the Nyquist plot might display a tail sloping at a 45-degree angle, indicating Warburg impedance (Zhang et al. 2023a). This phenomenon is associated with ionic diffusion processes within the perovskite material.

Nyquist plots are often employed in impedance spectroscopy, provide essential insights into the resistive and capacitive behavior of solar cell devices, particularly regarding charge transport and recombination mechanisms. The simulated Nyquist plot reflects the impedance characteristics of the proposed device structure and how these relate to the conduction band offset (CBO) and valence band offset (VBO) (Kumar et al. 2022). In the high-frequency region of the Nyquist plot, the semicircle typically corresponds to charge transport resistance and the capacitance at hetero-interfaces, such as between the ETL and the absorber or the HTL and the absorber (Prochowicz et al. 2020). A significantly high CBO can create a substantial barrier for electron injection, leading to increased resistance and a larger semicircle in this region (Kaphle et al. 2020). In contrast, an optimal or minimal CBO would lower this resistance, resulting in a smaller semicircle that indicates more efficient charge transfer.

The low-frequency region of the Nyquist plot is generally associated with recombination resistance and the capacitance related to the bulk of the absorber and its interfaces (Parikh et al. 2022). A larger VBO may present a barrier to hole extraction, which could increase recombination losses and, consequently, enlarge the semicircle in the low-frequency regime (Arockiya Dass et al. 2024). This behavior might also result in a downward shift of the Nyquist plot, reflecting reduced device efficiency. Conversely, a well-matched VBO that facilitates efficient hole extraction would produce a smaller semicircle, indicative of lower recombination losses. Overall, the shape and size of the semicircles in the Nyquist plot are influenced by the interplay between the CBO and VBO. An imbalance in these band offsets can lead to poor charge extraction and increased recombination, manifested as larger semicircles and lower efficiency in the device. On the other hand, optimizing the CBO and VBO to promote efficient charge extraction and minimize recombination can result in smaller semicircles and enhanced device performance (Sekar et al. 2023). By carefully analysing the Nyquist plots in relation to band offset data, it is possible to assess whether the CBO and VBO are positively or negatively impacting the overall performance of the device structure (Sekar et al. 2023).

Therefore, analysing these features of the Nyquist plot furnishes researchers with insights into charge transfer efficiency at various interfaces within the solar cell, recombination processes occurring within the perovskite layer, and ionic diffusion phenomena

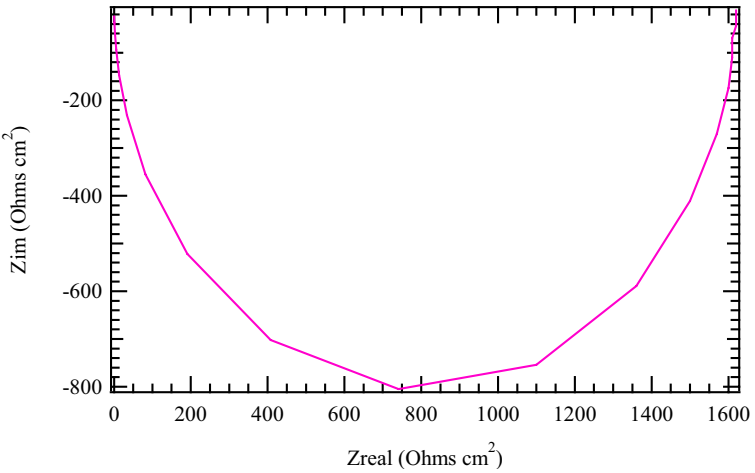



**Fig. 9** Nyquist plot of the proposed solar cell architecture

operating within the material. This continual analysis and optimization serve to advance the efficacy and reliability of PSCs, thereby contributing to the evolution of sustainable energy technologies. Impedance, which measures resistance to alternating current flow, comprises a real part (resistance) and an imaginary part (reactance). The horizontal axis, labelled Zreal (Ohms $cm^2$), represents the real component, while the vertical axis, labelled Zim (Ohms $cm^2$), represents the imaginary component. These labels likely denote the real and imaginary aspects of impedance, respectively, with both axes measured in $\Omega \cdot cm^2$. Generally, Fig. 9 depicts a semi-circle which implies effective charge transfer in the model cell architecture. This behaviour shows that the model cell device under study has robust electrical and optoelectronic capabilities (Korir et al. 2021).

### 3.9 The band alignment diagram of the model solar cell device

The lowest energy needed to excite an electron in a semi-conductor to a higher energy state is known as the band gap. A material can only absorb photon whose energy is greater than or equal to its band gap (Louis et al. 2023). Power is produced by a solar cell using the product of voltage and current. For PSCs to have a high PCE, the energy bands between the absorber and the surrounding charge extraction layers must be optimally aligned (Chaudhary et al. 2023). By locating the interface at which the photovoltaic cell inside the PSC develops, it is fundamentally interesting to examine the provided band energy diagram in order to advance the PCE. Finding methods to enhance PSCs will be made easier with an understanding of the device physics, particularly the band alignment across the interfaces (Bagade et al. 2023). With this information, new charge extraction layers may be developed to create appropriate interfaces with the perovskite absorber, maximizing the PCE and $V_{oc}$. Significant efforts have been made to date to increase the PCE by utilizing different electron extraction layer (EELs) and hole extraction layer (HELs) or by switching the metal contacts; however, because of a lack of understanding of the energy band diagram, these efforts have frequently been performed empirically without the benefit of universal guidelines (Hauff and Klotz 2022). Figure 9 clearly shows that the band gap energy changes with position and that the suggested solar cell's band alignment is not uniform. Surface potential

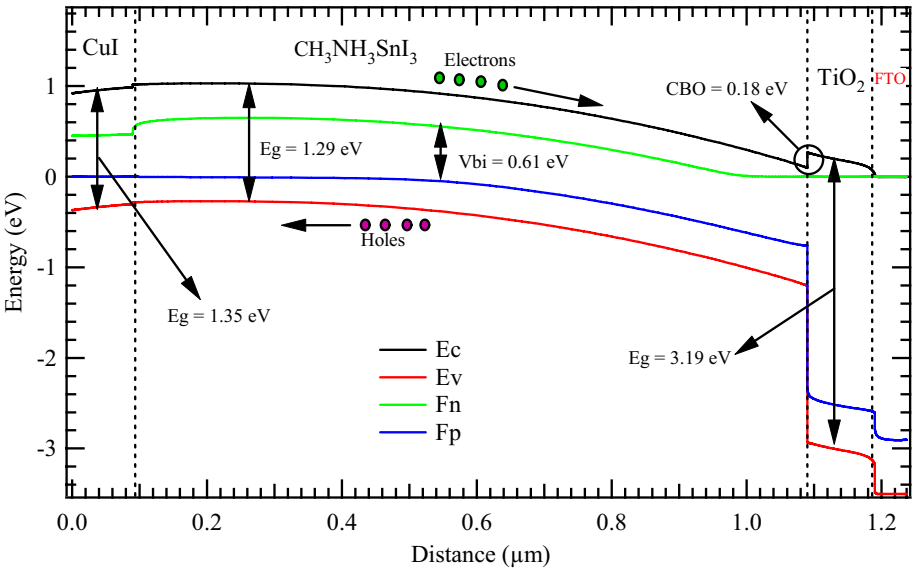



**Fig. 10** Band diagram for the proposed solar cell configuration

mapping and nanoscale ionic strain provide evidence that local ion accumulation as well as downward band bending can promote charge separation but impede charge collection at grain boundaries.

Figure 10 depicts the electronic structure of model solar cell architecture investigated in this work. The horizontal axis, measured in μm, tracks the distance across the interface where these two materials meet. The vertical axis, in electron volts (eV), represents the energy levels available for electrons to occupy. The shaded areas depict the energy range teeming with electrons, while the bands themselves map out the permitted energy levels. One key feature is the dramatic "spike" at the interface. This arises because the perovskite's conduction band, where electrons can readily flow, sits at a higher energy level compared to CuI conduction band. Furthermore, the entire energy landscape of the perovskite lies above that of CuI. This energy difference creates a driving force for electrons to migrate from the perovskite to CuI.

The Fermi level (Fn and Fp) is another crucial concept. It represents the energy level where the probability of finding an electron is exactly 50%. Perovskite, under illumination in a solar cell, deviates from perfect equilibrium, and the diagram depicts quasi-Fermi levels for both electrons (Fn) and holes (Fp) (Mabvuer et al. 2023). Holes are essentially the absence of electrons, and their movement also contributes to electrical current. Finally, the diagram highlights two crucial bands: the valence band and the conduction band (Bitton and Tessler 2023). The valence band houses the outermost electrons (valence electrons) of the material's atoms. The conduction band, on the other hand, is initially empty at absolute zero temperature but becomes crucial for conduction when electrons are excited. The band structure of a material significantly impacts its electrical properties, like conductivity and light absorption. In solar cells, understanding this electronic landscape is essential in optimizing how efficiently light energy is converted into electricity.

From discussion on Fig. 10 band diagram of a PSC provides an essential visual framework for understanding the energy levels and band bending within the device, which are

crucial for its operation. One of the key parameters depicted is the built-in potential (Vbi), which in this case is 0.61 eV. This potential is vital for driving the separation of photogenerated electrons and holes. A higher built-in potential generally enhances the efficiency of charge separation, reducing recombination losses. Thus, a Vbi of 0.61 eV indicates a strong driving force for charge separation and collection, which is favourable for the solar cell's performance.

Another critical parameter is the bandgap (Eg) of the materials involved. The perovskite layer has a bandgap of 1.29 eV, while $TiO_2$, used as the electron transport layer, has a bandgap of 3.19 eV. The bandgap of a material determines the range of the solar spectrum it can absorb. The perovskite's bandgap of 1.29 eV is well-suited for capturing a significant portion of the visible light spectrum, which is crucial for effective solar energy conversion. $TiO_2$ has a larger bandgap hence minimizing it from absorbing visible light, allowing it to efficiently transport electrons without competing for light absorption.

The conduction band offset (CBO) between the perovskite and $TiO_2$ layers is another important factor, measured at 0.18 eV. A positive CBO at this interface facilitates the injection of electrons from the perovskite into the $TiO_2$ layer. This positive offset is beneficial because it reduces the likelihood of recombination at the interface, ensuring that more electrons are transferred to the electron transport layer rather than recombining within the perovskite layer. Overall, the band diagram reveals that the positive CBO aids in efficient electron injection, the optimal bandgap of 1.29 eV enhances light absorption, and the built-in potential (Vbi) of 0.61 eV provides a substantial driving force for charge separation. These factors collectively contribute to the high efficiency of the solar cell. Figure 10 has no cliff-like features—valence band offset (VBO). This observation does not affect the device performance because the VBO)at the hetero-interface does not play a primary role provided the band gap is considerably larger than that of the perovskite (Crovetto and Hansen 2017).

However, several additional considerations must be addressed to optimize performance further. The presence of interface states at the perovskite-$TiO_2$ boundary can act as recombination centres, potentially reducing the device's efficiency. The mobility of charge carriers, both electrons and holes, within the perovskite and $TiO_2$ layers also affects the efficiency of charge transport. Finally, the overall architecture of the device, including layer thickness and additional components, plays a significant role in determining the solar cell's performance and stability. By carefully considering these aspects, researchers can refine the design and fabrication processes of perovskite solar cells, aiming for higher efficiencies and enhanced long-term stability.

To assist in determining whether interfaces have a built-in potential in the dark and a photovoltaic produced under illumination, Kelvin probe force microscopy (KPFM) as well as electron-beam induced current (EBIC) experiments are conducted (Hellmann et al. 2020). The position of the Fermi level in the energy gap of the various layers and the electrical structure, including the band line up and band bending at the interfaces, can only be inferred from the fluctuation of the work function over the entire device in KFPM measurements.

### 3.9.1 Effect of different buffer layer on the band diagram of the model cell

The buffer layer, which sits between the absorber and the charge transport layers, affects how well the energy levels are aligned, influencing charge extraction and recombination processes (Karthick et al. 2021). For example, using a CdS, ZnS and ZnSe buffer layers can enhance exciton extraction by aligning its conduction band edge favourably with that of the

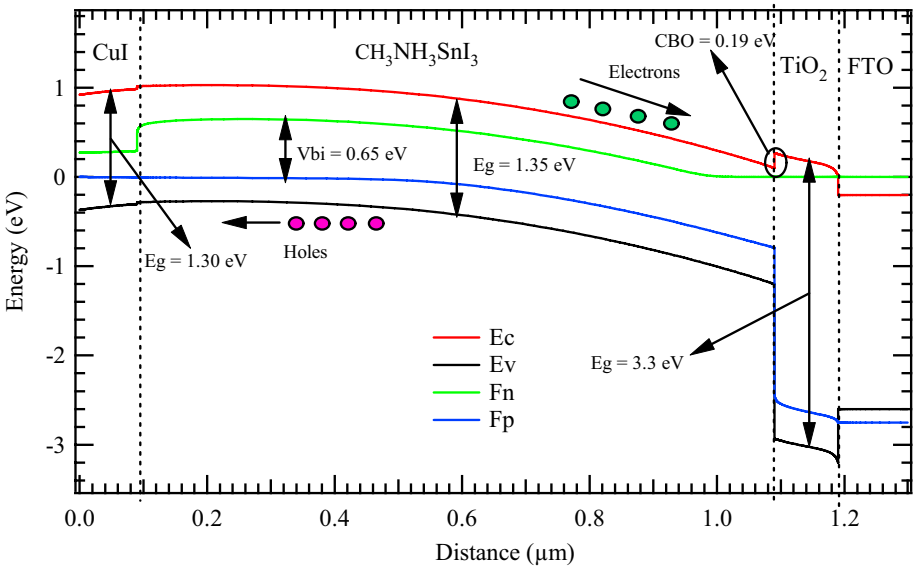



**Fig. 11** Impact of incorporating CdS buffer layer on the band diagram

perovskite, which can reduce recombination losses and improve efficiency (Njema et al. 2025). However, challenges such as stability and toxicity of CdS must be managed through material optimization and passivation engineering. CdS generally has a higher conduction band edge, which can better match the perovskite's conduction band, potentially enhancing electron extraction and reducing recombination losses (Lv et al. 2021). ZnS can also improve performance and stability by offering a favorable band alignment with certain perovskite materials. Thus, selecting the right buffer layer is crucial for optimizing the band diagram and, ultimately, the efficiency and stability of PSCs (Mourched et al. 2023).

Introducing a CdS buffer layer as shown in Fig. 11 has a substantial impact on the CBO by altering band alignment and interface properties. Typically, adding a CdS buffer layer results in a type-II band alignment at the interface with the absorber layer. In this setup, the valence band of the absorber is positioned higher than that of the CdS layer. This type-II alignment creates a relatively larger CBO, which offers several advantages. For instance, the increased CBO acts as a barrier that prevents holes from recombining at the interface, thereby boosting device efficiency. Also, it improves electron collection from the absorber to the front contact, further enhancing device performance. While a CdS buffer layer can increase the CBO due to type-II band alignment, the exact effect on CBO can vary depending on factors like interface quality, the presence of interfacial states, and doping levels. Optimizing these aspects is crucial for maximizing the efficiency and performance of solar cells. Figure 11 indicates a Vbi of about 0.65 eV, calculated from the separation between the Fermi levels of CuI and $TiO_2$. The conduction band edge (Ec) of the $CH_3NH_3SnI_3$ perovskite is approximately −3.8 eV, and the valence band edge (Ev) is around −5.4 eV, suggesting a bandgap of approximately 1.6 eV. Generally, the VBO at is not of primary importance provided the bandgap of the buffer higher than that of the absorber (Crovetto and Hansen 2017).

The incorporation of a ZnS buffer layer in the model solar cell can lead to significant improvements in device performance, as indicated by the energy level diagrams in Fig. 12.

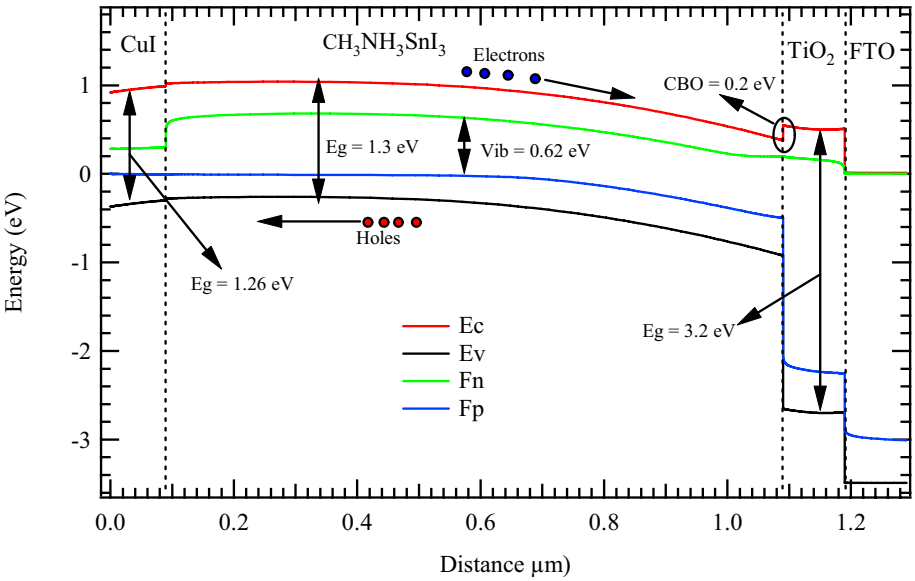



**Fig. 12** Impact of incorporating ZnS buffer layer on the band diagram

ZnS buffer layer, which has a bandgap of approximately 3.6 eV, provides several benefits due to its high energy bandgap compared to the 1.26 eV bandgap of $CH_3NH_3SnI_3$ absorber. This large bandgap helps protect the perovskite layer from potential damage caused by high-energy photons, ensuring that the underlying material remains intact and functional. In terms of band alignment, the conduction band of ZnS is positioned above that of the absorber.

Beyond performance, the ZnS layer contributes to the stability of the device. Its stable nature and wide bandgap offer protection to the $CH_3NH_3SnI_3$ photoactive layer, shielding it from environmental degradation and thus extending the lifespan of the solar cell. The key to optimizing this device lies in ensuring proper band alignment and minimizing interface states, which together enhance the device's performance and durability. The Vbi from Fig. 12 is around 0.62 eV. The separation between the valence band (Ev) and the conduction band is approximately 1.30 eV. The CBO at the $TiO_2/CH_3NH_3SnI_3$ interface is 0.20 eV which is ideal to facilitate effective charge separation at the interfaces, and prevents electrons from flowing back to the perovskite.

Figure 13, on the other hand, demonstrates how incorporating ZnSe buffer layer into a PSC impacts its performance by modifying the CBO and the overall bandgap (Eg) of the device (Roy and Khare 2022). When ZnSe is added, this offset can either increase or decrease, depending on the relative positions of the valence band edges. Accordingly, optimizing the CBO is essential for minimizing energy barriers and improving device efficiency by enhancing hole transport (Roy and Khare 2022). A well-matched CBO helps in efficiently transferring electrons from the perovskite to the ETL, which is crucial for maximizing the solar cell's efficiency. In this context, the CBO was found to be 0.22 eV which is still sufficient to allow the electrons to flow to the ETL without flowing back to the absorber to cause recombination, and device efficiency loss.

The presence of ZnSe can create a bandgap gradient at the interface with the perovskite. This gradient can enhance charge carrier collection by reducing recombination losses,

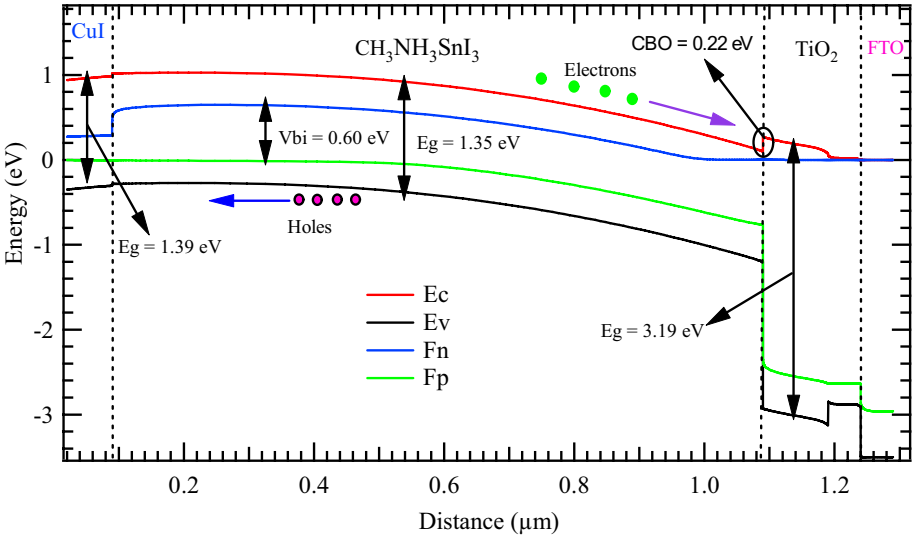



**Fig. 13** Impact of incorporating ZnSe buffer layer on the band diagram

thus improving device performance. Adding a ZnSe buffer layer can significantly affect the CBO and Eg of a perovskite solar cell. By adjusting the ZnSe layer's bandgap, thickness, and interface quality, these parameters can be optimized to boost the device operational performance. Key factors include achieving proper bandgap alignment, controlling the ZnSe layer thickness to manage band bending and offsets, and ensuring a high-quality interface to enhance charge carrier movement and reduce recombination. These considerations collectively contribute to the improved efficiency of perovskite solar cells with a ZnSe buffer layer. Generally, addition of buffers—CdS, ZnS and ZnSe increased the CBO marginally, 0.19, 0.20 and 0.22 eV, respectively. The model cell structure with the buffer gave a CBO of 0.18 eV.

## 4 Introduction of buffer layers on the model cell

In recent years, PSCs have attracted considerable attention due to their potential to achieve high efficiency coupled with low-cost fabrication processes. A crucial element within solar cell structures is the incorporation of buffers, which play a pivotal role in improving device performance and stability (Li et al. 2024). Buffers serve multifaceted functions within PSCs, including facilitating charge transport, aligning work functions, and passivating surface defects (Xu et al. 2023). First, buffers aid in the efficient transport of charge carriers, namely electrons and holes, within the device. By minimizing recombination losses and enhancing the collection of photogenerated carriers, they contribute to optimizing device efficiency (Nowsherwan et al. 2023). Furthermore, buffers assist in aligning the energy levels between different layers within the solar cell stack (Valenta et al. 2024). This alignment is essential for ensuring efficient charge extraction and reducing energy losses at interfaces. Moreover, buffers can effectively passivate defects on the surface of the perovskite layer, thereby reducing non-radiative recombination and improving overall device performance.

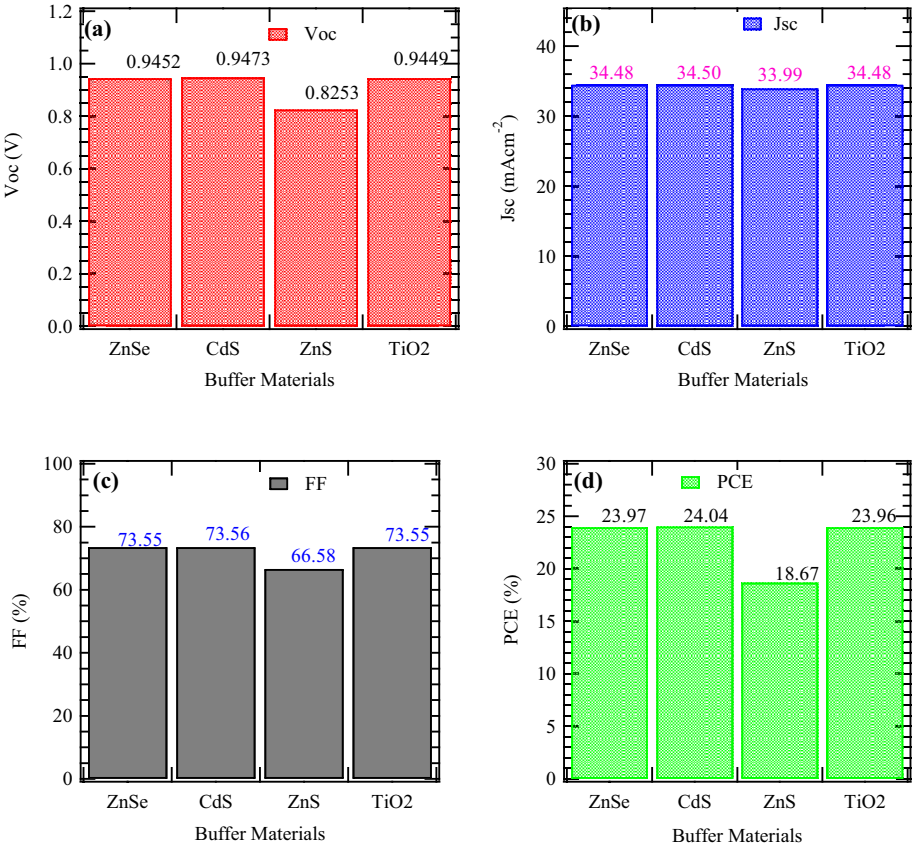



**Fig. 14** The effect on incorporating buffer materials on the model cell architecture

The materials commonly employed as buffers in PSCs vary depending on their intended function. For instance, n-type buffers, typically utilized on the ETL side of the device, include metal oxides such as zinc oxide (ZnO), titanium dioxide ($TiO_2$), and tin oxide ($SnO_2$) (Kim et al. 2023a). These materials aid in efficient electron transport and surface passivation. Conversely, p-type buffers, employed on the hole transport layer (HTL) side of the device, encompass organic materials such as poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) and inorganic materials like copper iodide (CuI) or nickel oxide (NiO), which facilitate efficient hole extraction and transport (Sajid et al. 2023b). Furthermore, certain materials can serve as dual-function interlayers, providing both electron and hole transport properties. An example of such materials is Spiro-OMeTAD, an organic compound capable of functioning as both a hole transport material and a passivation layer (Kim et al. 2023b). The selection of appropriate buffer materials hinges on various factors, including the device architecture, processing conditions, and desired performance metrics such as efficiency, stability, and scalability. Researchers are actively exploring novel buffer materials and device architectures to further enhance the performance and stability of perovskite solar cells.

The data presented in Fig. 14a showcases the $V_{oc}$ of different materials used in the proposed solar cell, indicating a substantial influence of buffer materials on Voc. The plot

demonstrates higher $V_{oc}$ values buffer layers such as ZnSe, CdS, ZnS and $TiO_2$. ZnS buffer, however, gave a low $V_{oc}$ compared to the other buffer materials. Extracted $V_{oc}$ values show that CdS buffer layer gave a better $V_{oc}$ of 0.9473 V as compared to 0.8762 V for the model cell without the buffer, which is a significant increase of ~ 8.11%. The mechanisms behind this observation include surface passivation, reducing electron–hole recombination, and improving electrical contact between cell layers, thereby enhancing current flow. The choice of buffer material depends on factors like the absorber layer's bandgap and metal contact's work function. Overall, integrating a buffer layer emerges as a promising strategy to boost solar cell performance. Figure 14b demonstrates the $J_{sc}$ of the solar cell model employing various buffer materials, a critical factor which influences solar cell efficiency. CdS depicts the highest $J_{sc}$ (34.50 mA/cm$^2$) among the materials tested, followed by CdS, ZnS, and $TiO_2$. This implies that integrating CdS as a buffer material may yield better $J_{sc}$ which results in a current enhancement of about 0.5%. It is crucial to acknowledge that $J_{sc}$ can be influenced by factors beyond buffer materials, such as the thickness of the buffer layer, properties of the absorber layer, and the overall solar cell design.

Figure 14c shows the impact of integrating buffer materials on FF. The vertical axis represents %FF, while the horizontal axis enumerates various materials – ZnSe, CdS, ZnS, and $TiO_2$. The data suggests a significant increase in FF with the addition of buffer materials. Notably, ZnS exhibits the lowest FF percentage at 66.58%, whereas $TiO_2$ demonstrates the highest FF at 73.56%. On the other hand CdS and ZnSe gave FF values of 73.55 and 73.55%, respectively. On the other hand, Fig. 14d depicts the influence of various buffer materials on the PCE of the model solar cell. It reveals a discernible trend indicating that the integration of buffer materials enhances the PCE of the solar cell structure, with CdS emerging as the most effective buffer among the tested materials, achieving a notable PCE of 24.04% which is enhancement of about 0.3% as compared to the model cell without a buffer. ZnS buffer material exhibited the lowest PCE of 18.67%. Generally, the observations noted from Fig. 10 underscores the critical role of buffer materials in enhancing solar cell performance by minimizing energy loss and optimizing charge carrier transport within the cell structure. The strategic incorporation of buffer materials holds promise in mitigating issues like carrier recombination, thereby improving overall solar cell efficiency, and may be critical in enhancing solar cell stability and durability.

The prominence of ZnSe and CdS as the most efficient buffer materials suggests their potential for adoption in solar cell technologies. Their modest performance is indicative of favorable characteristics such as appropriate energy band alignment and efficient charge transfer properties. This underscores the importance of ongoing research endeavors aimed at exploring and optimizing the utilization of buffer materials to enhance the efficiency and reliability of solar cell devices. In summary, the findings presented in Fig. 10d highlight the positive impact of buffer materials on solar cell PCE, with CdS particularly standing out as a promising buffer material. This emphasizes the pivotal role of material science and engineering in driving advancements in solar cell technology towards better operational efficiency and sustainability.

### 4.1 Effect of donor density

According to Fig. 15a, an increase in donor density leads to a rise in $J_{sc}$ but the value of $J_{sc}$ remains constant between $10^{10}$ and $10^{12}$ cm$^{-3}$ followed by a steady increase from $10^{12}$ to $10^{16}$ cm$^{-3}$ and later decreases before levelling off. Donor density represents the concentration of impurities contributing electrons to a semiconductor's conduction band. In PV

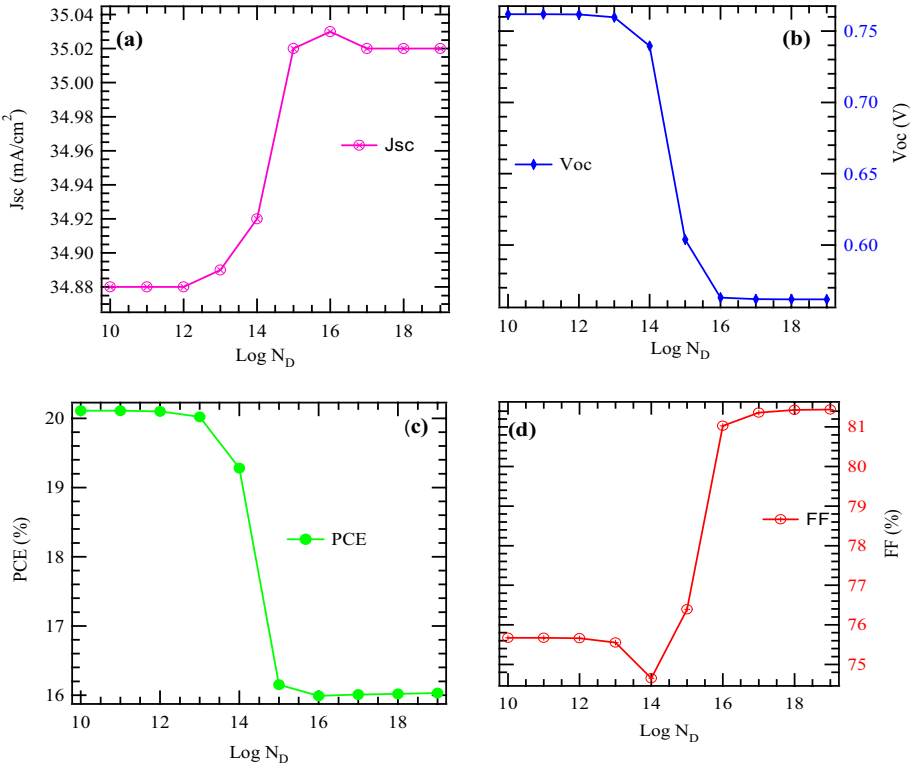



**Fig. 15** The effect of donor density on **a** $J_{sc}$, **b** $V_{oc}$, **c** PCE and **d** FF

devices, these extra electrons can enhance the current flow upon light absorption. Therefore, a higher donor concentration logically correlates with increased short-circuit current. Nevertheless, an optimal donor density likely exists. Excessive donors may hinder carrier mobility due to impurity interactions and facilitate recombination, ultimately diminishing $J_{sc}$ (Fig. 15a).

It can be observed from Fig. 15b that the value of $V_{oc}$ remains constant between $10^{10}$ and $10^{14}$ $cm^{-3}$ before it decreases with increase in donor density before levelling of between $10^{16}$ and $10^{19}$ $cm^{-3}$. On the other hand, Fig. 15c shows that PCE remains constant reaching an optimal value of 22.11% at $10^{10}$ $cm^{-3}$ and then decreases steadily with increase in doping concentration. PCE remains constant between $10^{10}$ and $10^{14}$ $m^{-3}$ followed by a slight decrease between $10^{14}$ and $10^{16}$ $cm^{-3}$ before levelling off while FF (Fig. 15d) remains steady between $10^{10}$ and $10^{13}$ $cm^{-3}$ before dropping slightly at $10^{14}$ $cm^{-3}$, and finally increases steadily between $10^{15}$ and $10^{19}$ $cm^{-3}$.

## 4.2 The effect of acceptor density

In Fig. 16a, $V_{oc}$ remains constant between $10^{10}$ to $10^{15}$ $cm^{-3}$ before attaining a steady increase from $10^{15}$ to $10^{17}$ $cm^{-3}$, then a decrease between $10^{17}$ a$10^{18}$ $cm^{-3}$, followed by a slight increase between $10^{18}$ and $10^{19}$ $cm^{-3}$. The optimum $V_{oc}$ is observed at $10^{17}$ $cm^{-3}$. Figure 16b $J_{sc}$ remains relatively constant between $10^{10}$ and $10^{14}$ $cm^{-3}$ followed by a

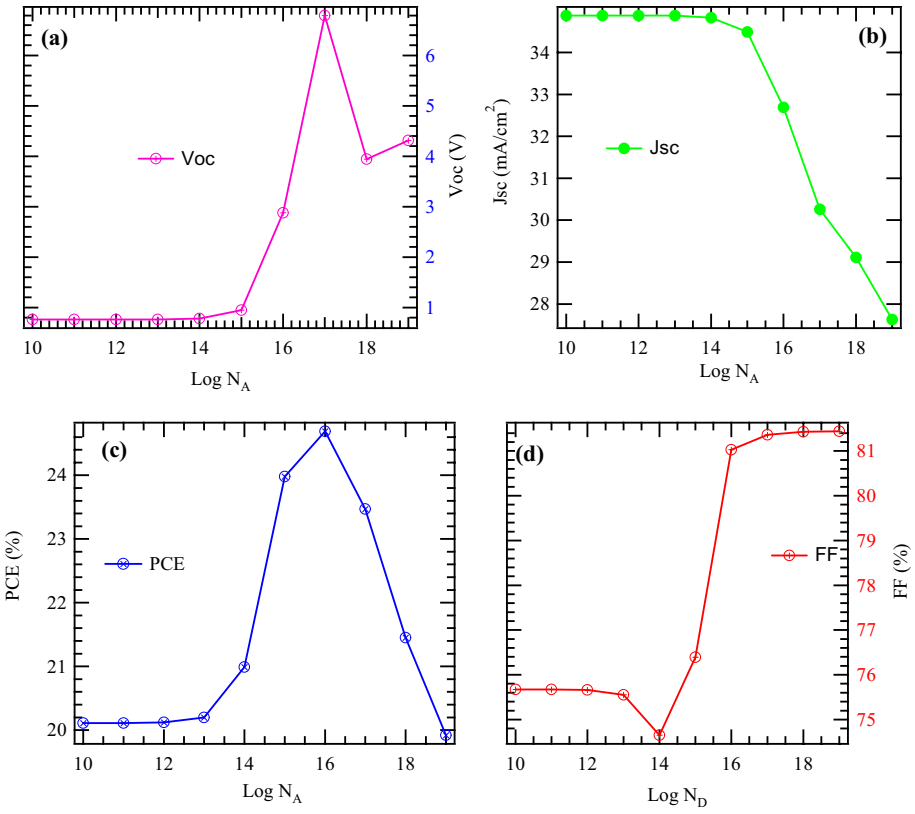



**Fig. 16** The effect of acceptor density on **a** $V_{oc}$, **b** $J_{sc}$, **c** PCE and **d** FF

decrease between $10^{15}$ and $10^{19}$ $cm^{-3}$ with increase in acceptor density. The optimum $J_{sc}$ is 35.02 $mA/cm^2$ between $10^{10}$ and $10^{14}$ $cm^{-3}$. However, from Fig. 16c, PCE remains apparently constant between $10^{10}$ and $10^{13}$ $cm^{-3}$ and then increases steadily between $10^{13}$ and $10^{16}$ $cm^{-3}$. PCE reaches an optimal value at $10^{16}$ $cm^{-3}$. Hence, there exists an optimal acceptor density balancing efficient hole extraction, minimizing recombination, and maintaining good hole mobility. Achieving this balance is critical for enhancing solar cell performance. Figure 16d shows the relationship between acceptor density in the absorber layer of the model solar cell and its impact on FF. Initially, increasing acceptor density enhances FF by improving hole extraction, reducing recombination and leakage currents. It is evident from the Fig. 16d that FF remains relatively constant between $10^{10}$ and $10^{13}$, and then dips between $10^{13}$ and $10^{14}$ $cm^{-3}$ before increasing steadily between $10^{14}$ to $10^{17}$ $cm^{-3}$ and ultimately achieves a maximum value between $10^{16}$ and $10^{19}$ $cm^{-3}$. The optimum FF is approximately 81.5% at $10^{19}$ $cm^{-3}$.

The proper functioning of a solar cell majorly relies on doping concentration along with defect densities. In this work, acceptor doping concentration is varied from $10^{10}$ to $10^{19}$ $cm^{-3}$ in order to observe the impact of doping on the electrical characteristic of the model solar cell. Doping concentration has significant electrical outcomes between $10^{15}$ and $10^{19}$ $cm^{-3}$ whereby the values of FF and $V_{oc}$ increases significantly. On the other hand,

PCE and $J_{sc}$ are negatively affected with increase in the doping concentration because of their observed decrease with increasing doping concentrations.

### 4.3 Effect of defect density

$MASnI_3$ exhibits promising qualities for utilization in PSCs owing to its favorable optical and electronic characteristics (Haneef et al. 2024). Nonetheless, similar to many emerging materials, $MASnI_3$ encounters challenges, with defect density emerging as a crucial factor impacting its functionality. Defects embedded within the crystalline structure of $MASnI_3$ can significantly influence its electronic attributes, including carrier mobility, recombination rates, and overall stability (Ayaydah et al. 2023). For instance, point defects such as vacancies or interstitials may serve as sites for charge trapping, resulting in heightened carrier recombination and reduced device efficiency (Zhang et al. 2023b). Furthermore, structural imperfections like grain boundaries or dislocations can hinder charge transfer within the material, further diminishing device performance (Srivastava et al. 2023). Perovskite solar cells (PSCs) are negatively impacted by defects in the absorber layer (Jan and Noman 2022). Even if they are undesirable, they cannot be avoided and have a significant impact on the device's functionality. The fluctuation of the performance parameters versus donor density is displayed in Fig. 10. Since carriers become trapped in the deep and shallow energy levels that the defects introduce, an increase in defect density also suggests an increase in non-radiative recombination of charge carriers.

Accordingly, understanding and regulating defect density in $MASnI_3$-based PSCs is imperative to achieve optimal device performance. Approaches to mitigate defect formation and density encompass meticulous adjustment of fabrication processes, such as precursor composition, deposition methods, and post-treatment techniques. Additionally, integrating passivation layers or conducting interface engineering can aid in diminishing defect-related recombination losses and enhancing device stability. Figure 17 shows that high defect density levels lead to decrease in the electrical parameters of the model cell structure. This phenomenon arises from defects within the solar cell material, which have the capability to ensnare charge carriers, thus impeding the current flow through the cell resulting in the decrease of cell metrics. The specific relationship between defect density and FF/PCE hinges on the material composition and design nuances of the solar cell. Nonetheless, a prevailing trend emerges where increased defect densities generally coincide with diminished FF and PCE. Accordingly, marginal alterations in defect density at elevated defect levels exert a more pronounced influence on FF and PCE compared to equivalent alterations at lower defect levels.

From Fig. 17a, $V_{oc}$ changes with varying defect density. As defect density increases, $V_{oc}$ remains relatively constant from $10^{11}$ to $10^{14}$ $cm^{-3}$, and then decreases as the defect density increases, reaching a value of 0.9452 V at $10^{18}$ $cm^{-3}$. On the other hand, Fig. 17b shows the relationship between $J_{sc}$ and defect density. The horizontal axis represents the defect density ($N_t$) of the light intensity. As the defect density increases, the $J_{sc}$ remains constant between $10^{11}$ and $10^{15}$ $cm^{-3}$ before decreasing steadily as the defect density rises from $10^{15}$ to $10^{18}$ $cm^{-3}$. From Fig. 17c, the PCE remains constant, giving a value of 23.98% between $10^{11}$ and $10^{15}$ $cm^{-3}$ before decreasing steadily with further increase in defect density. Figure 17d shows the relationship between FF and defect density. As the defect density increases, the FF remains largely constant between $10^{11}$ and $10^{15}$ $cm^{-3}$ before decreasing steadily as the defect density rises from $10^{15}$ to $10^{18}$ $cm^{-3}$. The optimal FF is 75.27% at $10^{15}$ $cm^{-3}$.

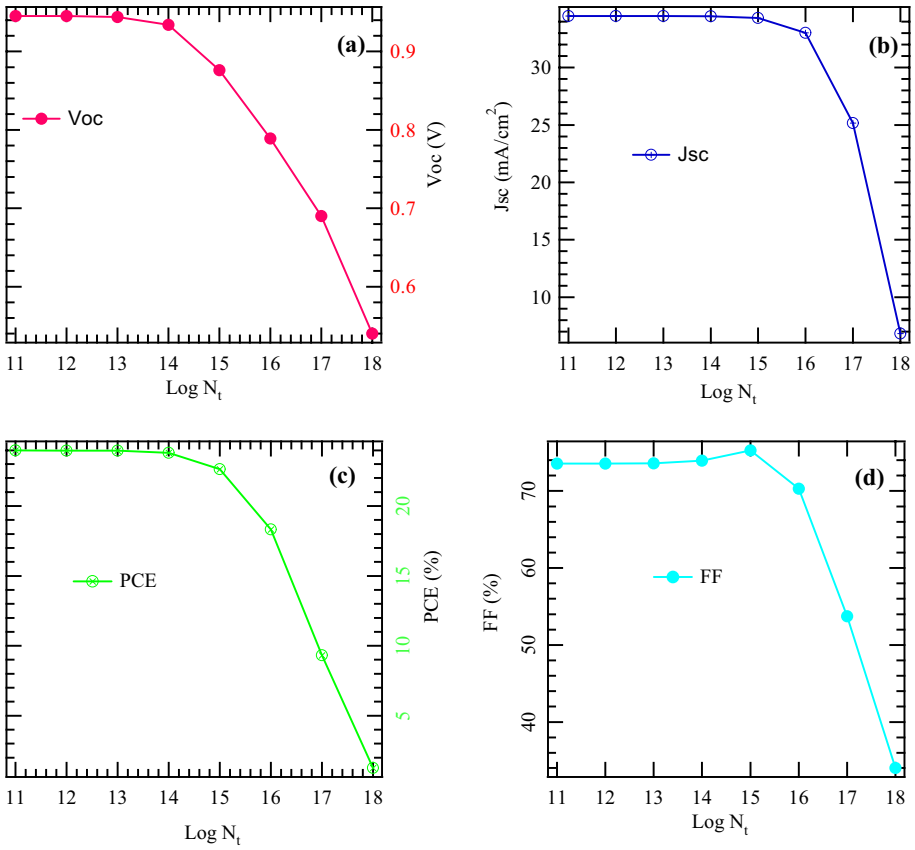



**Fig. 17** The effect of defect density on **a** $V_{oc}$, **b** $J_{sc}$, **c** PCE and **d** FF

## 5 Carrier lifetime and charge carrier diffusion length analysis

In PSCs, the charge carrier diffusion length plays a critical role in determining how effectively electrons and holes move through the absorber layer before recombining (Sekar et al. 2024). Diffusion length is highly dependent on the absorber thickness and defect density (Jan and Noman 2022). In thinner absorber layers, charge carriers can more easily reach the electrode contacts, minimizing recombination losses and boosting device efficiency (Sekar et al. 2024). However, excessively thin absorbers may not absorb enough light, which lowers the photocurrent (Tang et al. 2014). Conversely, thicker absorbers require charge carriers to travel longer distances, increasing recombination chances, especially if the diffusion length is shorter than the thickness. Defects in the material, particularly at grain boundaries, further shorten the diffusion length by serving as recombination centers, thereby limiting device performance (An et al. 2021).

For PSCs, the diffusion length and carrier lifetime are affected by variables like absorber layer thickness and defect density (Hasan et al. 2023). The carrier diffusion length ($L_D$) and lifetime ($\tau$) can be described using Eq. 8.

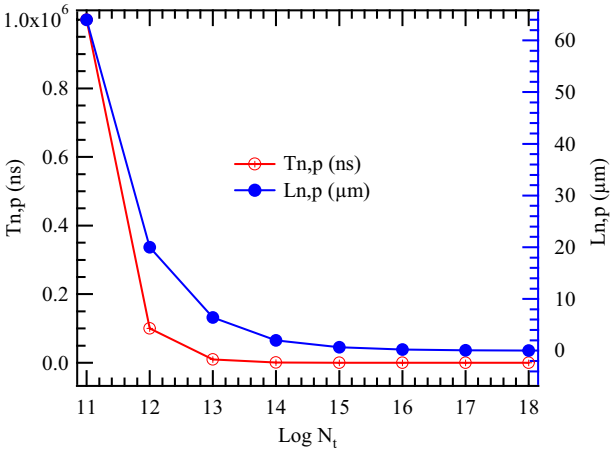



**Fig. 18** Carrier life time (τ) and diffusion length (L) plots of the model cell structure

$$L_D = \sqrt{D.T} \tag{8}$$

Here, D represents the diffusion coefficient, and ($\tau$) is the carrier lifetime. The diffusion length extends with increased carrier lifetimes and diffusion coefficients, which are determined by the defect density and film quality. Decreasing defect densities improves diffusion length by reducing non-radiative recombination losses.

The carrier lifetime is affected by the defect density ($N_t$), and is typically represented by Eq. 9.

$$\tau = \frac{1}{N_t.v_{th}.\sigma} \tag{9}$$

In this context, ($v_{th}$) represents the thermal velocity of charge carriers, while ($\sigma$) denotes the capture cross-section of defect states. An increase in defect density leads to a reduction in carrier lifetime, as it enhances trap-assisted recombination.

Carrier lifetime is linked to how long charge carriers can exist before recombining, with longer lifetimes allowing carriers to travel greater distances, which is advantageous for thicker absorbers (Njema and Kibet 2023b). However, high defect densities within the material can shorten the carrier lifetime by creating more non-radiative recombination sites. This negatively impacts performance, especially in thicker absorber layers where carriers need more time to reach the contacts. To optimize both diffusion length and carrier lifetime, PSCs require careful control over absorber thickness and defect reduction. Methods such as grain boundary passivation, enhancing crystal quality, and fine-tuning the absorber thickness can significantly improve both parameters, leading to better charge transport and higher PCE (Imran et al. 2024).

Figure 18 represents characteristic analysis of carrier life and diffusion length for both the electrons and holes. In this cell model, both the carrier lifetime and diffusion length of the electrons and holes are the same. The steep decline at the beginning of the curve suggests that the carrier lifetime decreases significantly as the defect density of the absorber increases. This rapid drop indicates increased recombination, defects, or material inhomogeneities which limit the movement of charge carriers, leading to reduced efficiency. On the other hand, L, represents the diffusion length of excitons, which similarly declines gradually. Diffusion length refers to the distance excitons can travel before recombining

(Akel et al. 2023). Carrier diffusion length decreases gradually in the same manner as carrier lifetime, reaching an optimal value of approximately 5 μm before levelling off significantly, whereas $\tau$ attains an optimal value of 1000 ns. The optimal defect density was achieved at about $10^{14}$ $cm^{-3}$ for this model cell architecture; nonetheless, absorber thickness did not appear to have any impact on the results reported in Fig. 18. These values represent the best operating conditions for reducing recombination while preserving efficient charge carrier transport. The decrease in the Ln and Lp curve suggests that diffusion length is also affected by defect density. This parameter is affected by factors such as material quality or interface effects, which may cause a more moderate reduction in the distance carriers can travel. Generally, the initial steep reduction in in carrier lifetime ($\tau$) points to a rapid degradation of carrier mobility, which can be attributed to increased recombination or defects at interfaces. These trends are critical in understanding the performance limitations of a solar cell or semiconductor device, particularly in relation to material interfaces and absorber properties.

## 6 Comparative analysis

Various studies have investigated solar cell devices with CuI as the HTL and recorded varying electric metrics as can be observed in Table 3. Although the list is not complete, it provides an overview of the impact of CuI on cell performance. Evidently, the experimental models presented herein, show considerably low performance outcomes largely attributed to defects at interfaces that may cause charge recombination losses, therefore resulting in energy losses and low electrical outcomes.

On the other hand, simulated cell devices show better electrical performance metrics because the simulations are conducted under ideal conditions. Numerical simulations give insights on promising material combinations and engineering parameters such as bandgap, defect density, acceptor and donor densities for better fabrication of solar cell structures. Clearly from Table 3, the model cell structure proposed in this work yields remarkable performance metrics as compared to other configurations. This may be attributed to meticulous optimization of material combination, thickness optimization of materials and intrinsic engineering parameters at interfaces. Therefore, the current cell model inspires further investigation especially in the experimental fabrication of a physical cell of robust operational performance.

## 7 Conclusions

This study found that temperature increase did not affect the electrical outcomes of the model cell considerably; implying that CuI hole transport layer employed in the cell architecture has an impressive temperature tolerance and better stability, maintaining stable electrical performance across a wide temperature range from 260 to 500 K. This characteristic property is crucial for the practical deployment of the model cell architecture in diverse environments, and accordingly inspires durability and reliability. This work also highlights the importance of optimizing defect densities and acceptor levels, achieving an ideal defect density of $10^{15}$ $cm^{-3}$ and an optimal acceptor density of $10^{16}$ $cm^{-3}$ whereas the optimal thickness of the absorber was noted at 2500 nm. Palladium as a back contact gave promising electrical output compared to other back contact materials tested. This can be

**Table 3** Comparative analysis with related solar cell devices

| Device architecture | Method | $V_{oc}$ (V) | $J_{sc}$ (mA $cm^{-2}$) | FF (%) | PCE (%) | References |
|---|---|---|---|---|---|---|
| Glass/FTO/$TiO_2$/$CH_3CH_3PbI_3$/CuI/C | Expt | 0.78 | 16.70 | 0.57 | 7.50 | Sepalage et al. (2015) |
| Glass/ITO/$SnO_2$/$CH_3CH_3PbI_3$/CuI/C | Expt | 0.79 | 19.30 | 0.54 | 8.30 | Srivastava et al. (2022) |
| Glass/TCO/$TiO_2$/$CH_3CH_3PbI_3$/CuI/Au | Simul | 0.712 | 26.55 | 74.05 | 14.0 | Yadav et al. (2019) |
| ITO/$TiO_2$/$MASnI_3$/CuI/back contact | Simul | 0.898 | 33.38 | 54.19 | 16.08 | Sharma and Kaushik (2022) |
| FTO/TiO2/$CH_3NH_3SnI_3$/CuI/Pd | Simul | 0.8762 | 34.33 | 75.27 | 23.98 | This work |

attributed to its high work function, stability under ambient conditions, and low resistivity. The optimal electrical outcomes of the proposed solar cell structure were; PCE of 23.98%, FF of 75.27%, $V_{oc}$ of 0.8762 V and $J_{sc}$ of 34.33 mA/cm$^2$. Among the buffer materials tested on the model cell, CdS exhibited the highest $V_{oc}$ (0.9473 V) and PCE (24.04%), surpassing other materials like ZnSe which posted slightly lower values for both $V_{oc}$ (0.9452 V) and PCE (23.97%). The built-in voltage ($V_{bi}$) obtained from the band diagram was consistent with that obtained from Mott-Skottky capacitance analysis (+0.63 eV). This is a critical property in charge transfer at interfaces. Analysis of carrier diffusion length and carrier lifetime yielded optimal values of 5 μm and 1000 ns, respectively. Ultimately, this work offers promising insights for the fabrication of a potentially cost-effective model cell structure based on CuI hole transport layer.

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**Authors' contributions** GGN: setting up numerical simulations, data analysis & editing, VK Simulation analysis and writing the first draft, BNM: installation and launching of SCAPS-1D and simulation analysis, JKK: method development, writing, editing, & supervision. All authors have read and approved the manuscript.

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## Declarations

**Conflict of interest** The authors declare no competing interests.

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**Consent for publication** This article has the consent of all the authors.

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